LeanForControl

Anand Gokhale Francesco Bullo

University of California, Santa Barbara

October 10, 2026

Chapter 1

Systems

The most general systems the library considers are $\dot{x} = F(t, x, u)$ in continuous time and $x^+ = F(k, x, u)$ in discrete time, each with an output $y = h(t, x, u)$. As for linear systems, there is one object per time domain: the data has the same shape, but a continuous trajectory is an integral curve and a discrete one is a recursion.

Special cases are constructors on the general object, each with a predicate recognising it: *time invariance* drops t, and *autonomy* drops u. Autonomy is used here in the control-theory sense, *without input*; much of the dynamical-systems literature uses the word for time invariance instead.

1.1 Continuous time

1.2 Discrete time

Chapter 2

Linear systems

2.1 The linear system objects

The data of a linear system — the four coefficient maps A, B, C, D — together with the equations that data satisfies. There are two objects, one per time domain, carrying the same API under the same names: a continuous-time trajectory is an integral curve, a discrete-time one is a sequence satisfying a recursion, and those are not two cases of one notion.

2.1.1 Continuous time

2.1.2 Discrete time

2.1.3 Time-invariant systems

A linear time-invariant system is a linear system whose coefficients do not vary with time, not an object of its own kind, so that every result about the time-varying object applies to it unchanged.

2.2 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \qquad y = Cx,$$

with state matrix A, input matrix B, and output matrix C, this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

2.2.1 Observability matrix

Definition 1 (Observability matrix)**.** The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n\times n}$ and $C \in \mathbb{F}^{p\times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C\,A \\ C\,A^2 \\ \vdots \\ C\,A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np)\times n}.$$

Rows are indexed by $\mathrm{Fin}\, n \times \mathrm{Fin}\, p$, so that A^k is available without casting $k : \mathrm{Fin}\, n$ through Fin.val.

Definition 2 (Observability)**.** A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C\,A^k\,x = 0 \qquad \text{for every } k = 0, 1, \ldots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge theorem 8 has real content.

Lemma 3 (Entries of the observability matrix)**.** *Block-row entry shape: at row (k, i) and column j, the observability matrix coincides with the (i, j) entry of $C\,A^k$:*

$$\mathcal{O}(A, C)_{(k,i),\,j} = (C\,A^k)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 1. □

Lemma 4 (Action of the observability matrix)**.** *For every state $x \in \mathbb{F}^n$ and every $(k, i) \in \mathrm{Fin}\, n \times \mathrm{Fin}\, p$,*

$$\left(\mathcal{O}(A, C) \cdot x\right)_{(k,i)} = \left(C\,A^k \cdot x\right)_i.$$

This is the definitional bridge that powers theorem 8.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see definition 1). □

2.2.2 Controllability matrix

Definition 5 (Controllability matrix)**.** The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = \begin{bmatrix} B & A\,B & A^2\,B & \cdots & A^{n-1}\,B \end{bmatrix} \in \mathbb{F}^{n\times(nm)}.$$

Columns are indexed by $\mathrm{Fin}\, n \times \mathrm{Fin}\, m$, so that A^k is available without casting $k : \mathrm{Fin}\, n$ through Fin.val.

Lemma 6 (Entries of the controllability matrix)**.** *Block-column entry shape: at row i and column (k, j), the controllability matrix coincides with the (i, j) entry of $A^k\,B$:*

$$\mathcal{C}(A, B)_{i,\,(k,j)} = (A^k\,B)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 5. □

Definition 7 (Controllability)**.** A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \ldots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k \, B \, u_k.$$

This phrasing does not name the controllability matrix, so the bridge theorem 10 has real content.

2.2.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 8 (Observability via a trivial kernel)**.** *A finite-dimensional linear system (A, C) is observable in the sense of definition 2 if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication:*

$$\text{IsObservable}(A, C) \iff (\forall x,\ \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0).$$

Proof. Both directions follow from the bridge lemma 4, which identifies the (k, i)-coordinate of $\mathcal{O}(A, C) \cdot x$ with the i-coordinate of $(C\,A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. □

2.2.4 Rank characterizations

Theorem 9 (Observability rank test)**.** *A finite-dimensional system (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e.*

$$\operatorname{rank} \mathcal{O}(A, C) = n,$$

where n is the state dimension.

Proof. Chain the kernel-form milestone theorem 8 with the matrix-level bridge $(\forall x,\ M \cdot x = 0 \Rightarrow x = 0) \iff \operatorname{rank} M = n$ from ‘MatrixAlgebra.Rank‘, applied with $M = \mathcal{O}(A, C)$. □

Theorem 10 (Controllability rank test)**.** *A linear system (A, B) is controllable if and only if the controllability matrix has full row rank,*

$$\operatorname{rank} \mathcal{C}(A, B) = n,$$

where n is the state dimension.

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y,\ \exists x,\ M\,x = y) \iff \operatorname{rank} M = n$ from ‘MatrixAlgebra.Rank‘, applied with $M = \mathcal{C}(A, B)$. □

2.2.5 Hautus observability

Definition 11 (Unobservable subspace)**.** The *unobservable subspace* of (A, C) is the A-invariant subspace

$$\mathcal{N}(A, C) \;=\; \{\, v \in \mathbb{C}^n \;:\; C\,A^k\,v = 0 \text{ for every } k = 0, 1, \ldots, n-1 \,\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge theorem 12 reads as content.

Theorem 12 (Observability via the unobservable subspace)**.** *A finite-dimensional system (A, C) is observable in the sense of definition 2 if and only if its unobservable subspace is trivial:*

$$\mathcal{N}(A, C) = \{0\} \iff \text{IsObservable}(A, C).$$

Proof. Both directions are membership unfoldings of definition 11 against definition 2. □

Lemma 13 (A-invariance of the unobservable subspace)**.** *The unobservable subspace is closed under the action of A: for every $v \in \mathcal{N}(A, C)$, also $A\,v \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $C\,A^n\,v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$-linear combination of $C\,A^i\,v$ for $i < n$. Each of those is zero, so the combination is zero. □

Definition 14 (Hautus observability matrix)**.** The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) \;=\; \begin{bmatrix} \mu I - A \\ C \end{bmatrix} \in \mathbb{C}^{(n+p)\times n}.$$

Lemma 15 (Kernel of the Hautus matrix)**.** *For every $\mu \in \mathbb{C}$ and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (A\,v = \mu\,v \;\wedge\; C\,v = 0).$$

Proof. □

Theorem 16 (Unobservability gives a Hautus rank drop)**.** *If (A, C) is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$*.*

Proof. The unobservable subspace definition 11 is nontrivial by theorem 12. It is A-invariant by lemma 13, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via lemma 15. □

Theorem 17 (A Hautus rank drop gives unobservability)**.** *If $\mu \in \mathbb{C}$ and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k, $A^k v = \mu^k v$, so $C\,A^k\,v = \mu^k\,Cv = 0$ for every k. The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. □

Theorem 18 (Hautus (PBH) test for observability)**.** *A finite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\text{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C},\ \ker H_{A,C}(\mu) = \{0\}.$$

Proof. Combine theorem 16 (failure direction) and theorem 17 (converse). □

2.2.6 Hautus controllability

Definition 19 (Hautus controllability matrix)**.** The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H^{\mathrm{ctrl}}_{A,B}(\mu) \;=\; [\mu I - A \quad B] \in \mathbb{C}^{n\times(n+m)}.$$

Theorem 20 (Hautus (PBH) test for controllability)**.** *A finite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H^{\mathrm{ctrl}}_{A,B}(\mu)$ has full row rank:*

$$\mathrm{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C},\ \operatorname{rank} H^{\mathrm{ctrl}}_{A,B}(\mu) = n.$$

Proof. Combine the duality bridge `isControllable_iff_isObservable_transpose` with the observability Hautus iff theorem 18, then convert the kernel-form RHS to the rank form via the matrix bridge `mulVec_kernel_trivial_iff_rank_eq_card_cols` from 'MatrixAlgebra.Rank', and finally identify the transposed Hautus matrices. □

2.3 Reachable subspace and Kalman decomposition

Definition 21 (Reachable subspace)**.** The reachable subspace of a pair (A, B) is the column span of its controllability matrix:

$$\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B).$$

Thus it is exactly the span of $B, AB, \ldots, A^{n-1}B$.

Reference: Hespanha, *Linear Systems Theory.*

Lemma 22 (Membership in the reachable subspace)**.** *Let $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$, and let $\mathcal{C}(A, B) \in \mathbb{F}^{n\times(nm)}$ be their controllability matrix. For $x \in \mathbb{F}^n$,*

$$x \in \mathcal{R}(A, B) \quad\iff\quad \exists u : \mathrm{Fin}(n) \times \mathrm{Fin}(m) \to \mathbb{F}, \qquad \mathcal{C}(A, B)u = x.$$

Here $\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B)$ is the reachable subspace.

Proof. □

Theorem 23 (Controllability via the reachable subspace)**.** *A pair (A, B) is controllable if and only if its reachable subspace is the whole state space.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Lemma 24 (A-invariance of the reachable subspace)**.** *The reachable subspace is A-invariant: $A\mathcal{R}(A, B) \subseteq \mathcal{R}(A, B)$.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Definition 25 (Kalman decomposition)**.** A Kalman decomposition of (A, B, C) records four coordinate sectors adapted to the reachable and unobservable subspaces. Their direct sums recover those subspaces and the full state space. The chosen complements are noncanonical and are not individually asserted to be invariant under A.

References: Hespanha, *Linear Systems Theory*; Kailath, *Linear Systems.*

Theorem 26 (Existence of the Kalman decomposition)**.** *Every finite-dimensional complex state-space system has four coordinate sectors adapted to its reachable and unobservable subspaces. The complements are noncanonical vector-space complements and are not individually asserted to be invariant under A.*

References: Hespanha, Linear Systems Theory*; Kailath,* Linear Systems*.*

Proof. □

Definition 27 (Adapted coordinate equivalence)**.** Addition of the four Kalman coordinate sectors defines a linear equivalence $X_{c\bar{o}} \times X_{co} \times X_{\bar{c}\bar{o}} \times X_{\bar{c}o} \simeq \mathbb{C}^n$. Choosing sector bases therefore gives an adapted basis of the original state space.

Theorem 28 (Block-zero pattern in adapted coordinates)**.** *In a basis adapted to the four Kalman coordinate sectors, ordered as controllable-unobservable, controllable-observable, uncontrollable-unobservable, uncontrollable-observable, the system has the forced-zero pattern*

$$A' = \begin{bmatrix} * & * & * & * \\ 0 & * & 0 & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}, \qquad B' = \begin{bmatrix} * \\ \\ 0 \\ 0 \end{bmatrix}, \qquad C' = \begin{bmatrix} 0 & * & 0 & * \end{bmatrix}.$$

Every displayed zero is asserted entrywise; no vanishing claim is made for the starred blocks. The chosen complement sectors are not individually asserted to be invariant under A.

References: Hespanha, Linear Systems Theory*; Kalman, "Mathematical Description of Linear Dynamical Systems" (1963); Kailath,* Linear Systems*.*

Proof. Reachable-subspace invariance forces the lower-left state and input zeros, unobservable-subspace invariance forces the remaining state zeros, and the zeroth observability condition forces the output zeros. Coordinate-basis representation lemmas translate these component statements to individual matrix entries. □

2.4 Hurwitz stability

For a real square matrix A, Hurwitz stability is expressed through the complex eigenpairs of its entrywise complexification, over an abstract index type. The rate-indexed predicate records a strict spectral margin. Ordinary Hurwitz stability is the zero-rate case, but is stated separately rather than derived, so that it unfolds directly to $\mathrm{Re}(\mu) < 0$.

Theorem 29 (Rate-Hurwitz via a spectral shift)**.** *A real matrix A is Hurwitz with decay rate α if and only if the spectrally shifted matrix $A + \alpha I$ is Hurwitz.*

Reference: João P. Hespanha, Linear Systems Theory *(2nd ed.), continuous-time stability criterion. This spectral-shift corollary is proved directly from eigenpairs.*

Proof. A complex eigenvalue μ of A becomes $\mu + \alpha$ after the shift, and $\mathrm{Re}(\mu) < -\alpha$ is equivalent to $\mathrm{Re}(\mu + \alpha) < 0$. □

The exponential decay estimates rest on a general Banach-algebra fact, stated with no matrices in it: an element of spectral radius below one has a contractive power.

Lemma 30 (A contractive power from a spectral radius below one)**.** *Let $\mathbb{A}$ be a nontrivial complete normed algebra over $\mathbb{C}$ and let $a \in \mathbb{A}$ have spectral radius $r(a) < 1$. Then there is an $m > 0$ with $\|a^m\| < 1$.*

Reference: Rudin, Functional Analysis *(Gelfand's spectral-radius formula).*

Proof. Gelfand's formula gives $\|a^m\|^{1/m} \to r(a)$, so $\|a^m\|^{1/m} < 1$ for all large m; pick any such m with $m \neq 0$. Were $\|a^m\| \geq 1$, raising to the positive power $1/m$ would give $\|a^m\|^{1/m} \geq 1$, a contradiction. □

2.5 Matrix Lyapunov equations and quadratic certificates

Definition 31 (Matrix quadratic form)**.** A real matrix P represents the quadratic form $x \mapsto x^{\mathsf{T}} P x$ on Euclidean state space.

Reference: standard quadratic Lyapunov-function construction.

Theorem 32 (Lyapunov equation for a Hurwitz matrix)**.** *If a real matrix A is Hurwitz, then for every positive-definite Q there is a positive-definite matrix P satisfying $PA + A^{\mathsf{T}} P = -Q$, and this P is the unique matrix solution.*

Reference: Khalil, Nonlinear Systems*.*

Proof. A contracting integer-time matrix exponential is obtained from the Hurwitz spectrum. Integrating over one time block and summing the resulting discrete Lyapunov series constructs P; contraction proves convergence and uniqueness. □

Theorem 33 (Quadratic instability certificate from an unstable eigenpair)**.** *A complex eigenpair of a real matrix A with $\operatorname{Re}\mu > 0$ yields a Hermitian real matrix H, a shift $\alpha > 0$, and a direction w such that $w^{\mathsf{T}} H w > 0$ and*

$$HA + A^{\mathsf{T}} H - 2\alpha H$$

is positive definite.

Reference: Hahn, Stability of Motion*; Khalil,* Nonlinear Systems*.*

Proof. Choose a shift avoiding the finitely many pairwise spectral resonances, solve the resulting Lyapunov–Sylvester equation by finite-dimensional injectivity, and extract w from the real or imaginary part of the supplied eigenvector. □

2.6 Peano–Baker series and the state transition matrix

For a genuinely time-varying state matrix $A(t)$, this section constructs the state transition matrix $\Phi(t, t_0)$ as the Peano–Baker series and proves Theorem 5.1: Φ solves the matrix ODE $\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0)$ with $\Phi(t_0, t_0) = I$, and consequently $x(t) := \Phi(t, t_0)x_0$ solves $\dot{x} = A(t)x$, $x(t_0) = x_0$.

Standing hypotheses. Throughout this section A is continuous and satisfies $\|A(s)\| \leq M$ for all s on the segment between the two times named in each statement. Both are needed: continuity to differentiate the series term by term, the bound to make it converge at all. Statements below do not repeat them.

Definition 34 (Peano–Baker term)**.** The k-th term of the Peano–Baker series for a time-varying state matrix $A(t)$:

$$P_0(t, t_0) := I, \qquad P_{k+1}(t, t_0) := \int_{t_0}^{t} A(s)\, P_k(s, t_0)\, \mathrm{d}s.$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5 (Peano-Baker series).

Definition 35 (State transition matrix)**.** The *state transition matrix* $\Phi(t, t_0)$, given by the Peano–Baker series

$$\Phi(t, t_0) := \sum_{k=0}^{\infty} P_k(t, t_0).$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Theorem 5.1 (Peano-Baker series).

Lemma 36 (Continuity of the Peano–Baker terms)**.** *Each term $P_k(t, t_0)$ of the Peano–Baker series is continuous in t, for A continuous.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Induction on k: P_0 is constant, and P_{k+1} is the primitive of a continuous integrand (definition 34), hence continuous by the fundamental theorem of calculus. □

Lemma 37 (Factorial bound on the Peano–Baker terms)**.** *If $\|A(s)\| \le M$ for all s between t_0 and t_1, then*

$$\|P_k(t, t_0)\| \;\le\; \|I\| \cdot \frac{(M|t - t_0|)^k}{k!} \qquad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Induction on k: bound the integrand $A(s)P_k(s, t_0)$ pointwise on the segment between t_0 and t, using submultiplicativity of the matrix norm and the induction hypothesis. Split on whether $t_0 \le t$ or $t \le t_0$ and integrate the majorant via the power rule, flipping the integral's orientation (`intervalIntegral.integral_symm`) in the backward case. □

Lemma 38 (Factorial bound on a subinterval)**.** *The bound in lemma 37, uniformized to the worst case $t = t_1$:*

$$\|P_k(t, t_0)\| \;\le\; \|I\| \cdot \frac{(M|t_1 - t_0|)^k}{k!} \qquad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Monotonicity of $t \mapsto (M|t - t_0|)^k$ on the segment between t_0 and t_1, combined with lemma 37. □

Lemma 39 (Summability of the Peano–Baker series)**.** *The Peano–Baker series $\sum_{k=0}^{\infty} P_k(t, t_0)$ converges absolutely, for A bounded between t_0 and t (in either order).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Compare against the exponential series $\sum_k (M|t - t_0|)^k / k!$ via lemma 37 and the comparison test. □

Lemma 40 (Continuity of the state transition matrix)**.** *$\Phi(\cdot, t_0)$ is continuous on the segment between t_0 and t_1 (including both endpoints), for A continuous and bounded by M there.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Each term $P_k(\cdot, t_0)$ is continuous (lemma 36) with a summable, point-independent bound (lemma 38); apply the Weierstrass M-test for continuity of a series. □

Theorem 41 (The state transition matrix solves the matrix ODE)**.** *The state transition matrix solves the matrix ODE*

$$\dot{\Phi}(t, t_0) = A(t)\,\Phi(t, t_0)$$

for t strictly between t_0 and t_1.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Differentiate the Peano–Baker series term by term, using lemma 37 as the uniform bound needed for term-by-term differentiation of a series. The $k = 0$ term is constant, so peel it off and match the shifted series' derivative against $A(t)\Phi(t, t_0)$ using linearity of the sum under left multiplication by $A(t)$. □

Theorem 42 (The state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Every term of the Peano–Baker series beyond $k = 0$ integrates over the degenerate interval $[t_0, t_0]$ and vanishes. □

Theorem 43 (Peano–Baker series: existence)**.** *Hespanha, Theorem 5.1. $x(t) := \Phi(t, t_0)\,x_0$ solves*

$$\dot{x}(t) = A(t)\,x(t).$$

Proof. Push theorem 41 through the fixed linear map $M \mapsto Mx_0$ (continuous, since the domain is finite-dimensional). □

Theorem 44 (Matrix-valued solutions)**.** *For a constant matrix C, $Y(t) := \Phi(t, t_0)\,C$ solves*

$$\dot{Y}(t) = A(t)\,Y(t).$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Right-multiply theorem 41 by the constant C (`HasDerivAt.mul_const`), then reassociate. □

Lemma 45 (Integral form, vector-valued)**.** *$x(t) := \Phi(t, t_0)\,x_0$ satisfies the integral equation*

$$x(t) = x_0 + \int_{t_0}^{t} A(s)\,x(s)\,\mathrm{d}s$$

for every t between t_0 and t_1.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. The fundamental theorem of calculus applied to theorem 43, using $\Phi(t_0, t_0) = I$ (theorem 42) to fix the initial value. □

Lemma 46 (Integral form, matrix-valued)**.** *For a constant matrix C, $Y(t) := \Phi(t, t_0)\, C$ satisfies the integral equation*

$$Y(t) = C + \int_{t_0}^{t} A(s)\, Y(s)\, \mathrm{d}s$$

for every t between t_0 and t_1.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. The fundamental theorem of calculus applied to theorem 44, using $\Phi(t_0, t_0) = I$ (theorem 42) to fix the initial value. □

2.6.1 Uniqueness

Theorem 47 (Peano–Baker series: uniqueness)**.** *Hespanha, Theorem 5.1. Any solution z of $\dot{x} = A(t)\, x$, $x(t_0) = x_0$, for t between t_0 and t_1, coincides with $x(t) := \Phi(t, t_0)\, x_0$.*

Proof. Apply the continuous-dependence bound of Theorem 3.4 with zero perturbation: since both solutions share the initial value x_0, the resulting bound on $\|x(t) - z(t)\|$ collapses to 0. □

Theorem 48 (Columns of the state transition matrix)**.** *Hespanha, P5.2. For every fixed i, the i-th column of $\Phi(t, t_0)$ is the unique solution of $\dot{x} = A(t)\, x$, $x(t_0) = e_i$, where e_i is the i-th standard basis vector.*

Proof. Restatement of theorem 47 at $x_0 := e_i$, using $\Phi(t, t_0)\, e_i = (\Phi(t, t_0))_{\cdot, i}$ (`Matrix.mulVec_single_one`). □

Theorem 49 (Uniqueness of matrix-valued solutions)**.** *Any solution Y of $\dot{Y} = A(t)\, Y$, $Y(t_0) = C$, for t between t_0 and t_1, coincides with $\Phi(t, t_0)\, C$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Apply the continuous-dependence bound of Theorem 3.4 with zero perturbation: since both solutions share the initial value C, the resulting bound on $\|Y(t) - Z(t)\|$ collapses to 0. □

2.6.2 Semigroup, invertibility and composition

Uniqueness is what makes these available: two expressions that solve the same matrix initial value problem must agree, and each identity below is an instance of that.

Theorem 50 (Semigroup property)**.** *Hespanha, P5.3. For A continuous and bounded on $[\tau, t]$, and any $\tau \le s < t$,*

$$\Phi(t, s)\, \Phi(s, \tau) = \Phi(t, \tau).$$

Proof. $\Phi(\cdot, \tau)$, re-anchored at s, is an integral solution of the matrix ODE on the segment between s and t with initial value $\Phi(s, \tau)$, hence coincides with $\Phi(\cdot, s)\, \Phi(s, \tau)$ there by theorem 49. □

Theorem 51 (Invertibility of the state transition matrix)**.** *Hespanha, P5.4. For A continuous and bounded on the segment between τ and t, $\Phi(t, \tau)$ is nonsingular and*

$$\Phi(t, \tau)^{-1} = \Phi(\tau, t).$$

Proof. From theorem 50, generalized to arbitrary order by re-anchoring $\Phi(\cdot, \tau)$ at t and invoking uniqueness there, $\Phi(\tau, t)\Phi(t, \tau) = \Phi(\tau, \tau) = I$, so $\Phi(t, \tau)$ and $\Phi(\tau, t)$ are mutual inverses. □

Lemma 52 (Composition on a subinterval)**.** *For every z on the segment between a and b,* $\Phi(z, a) = \Phi(z, b)\,\Phi(b, a)$. *Composition with the* first *argument free.*

Proof. $\Phi(\cdot, a)$, re-anchored at b, is an integral solution of $\dot{Y} = A(r)Y$ with value $\Phi(b, a)$ at b; so is $\Phi(\cdot, b)\Phi(b, a)$. They agree by theorem 49. □

Theorem 53 (Semigroup property, order-free)**.** *Hespanha, P5.3. For* any *s on the segment between τ and t,*

$$\Phi(t, s)\,\Phi(s, \tau) = \Phi(t, \tau).$$

The $\tau \le s < t$ ordering in theorem 50 matches the textbook's literal statement but is not a genuine requirement; the variation-of-constants proof needs the unordered form, where $s = t_0$ is fixed and τ ranges over the whole segment.

Proof. lemma 52 gives $\Phi(s, \tau) = \Phi(s, t)\Phi(t, \tau)$; left-multiply by $\Phi(t, s)$ and cancel using $\Phi(t, s)\Phi(s, t) = I$ (theorem 51). □

Lemma 54 (Change of base time)**.** *For every τ on the segment between t_0 and t,* $\Phi(t, t_0)\,\Phi(t_0, \tau) = \Phi(t, \tau)$*: composition through a fixed* base *point. This differs from theorem 53 in which of the three times is required to lie between the others — here the two fixed integration endpoints are the outer ones and τ is free, which is the shape variation of constants needs.*

Proof. theorem 53 with τ itself as the middle point gives $\Phi(t, \tau)\Phi(\tau, t_0) = \Phi(t, t_0)$; right-multiply by $\Phi(t_0, \tau)$ and cancel $\Phi(\tau, t_0)\Phi(t_0, \tau) = I$. □

Lemma 55 (Continuity in the initial time)**.** *With the first argument held fixed, $\tau \mapsto \Phi(t, \tau)$ is continuous on the segment between t and t_1 — the mirror of lemma 40, which varies the first argument. Needed wherever $\Phi(t, \cdot)$ is an integrand, as in variation of constants.*

Proof. By theorem 51, $\Phi(t, \tau) = \Phi(\tau, t)^{-1}$, so the map is $\tau \mapsto \Phi(\tau, t)$ — continuous in its first argument — followed by matrix inversion, which is continuous at every unit. □

2.6.3 Variation of constants

With a forcing term $B(t)u(t)$, the solution acquires a convolution against Φ. The difficulty is that the elapsed time appears both as the upper limit of integration and inside the integrand; the proof factors the two apart rather than differentiating Φ in its second argument.

Theorem 56 (Variation of constants: uniqueness)**.** *Hespanha, Theorem 5.2. Any two continuous integral solutions of the forced system $\dot{x} = A(t)x + B(t)u(t)$ with the same initial value x_0 agree on the segment between t_0 and t_1.*

Proof. For fixed t the right-hand side $v \mapsto A(t)v + B(t)u(t)$ is affine in v, hence Lipschitz with constant $\max(M, 1)$ where $\|A\| \le M$. Continuous dependence on initial conditions with zero perturbation then bounds $\|z_1(t) - z_2(t)\|$ by 0. The forcing term $B(t)u(t)$ is common to both solutions and cancels in the difference, so it never enters the Lipschitz estimate. □

2.6.4 The forced response as a trajectory

The solution theory above is stated on the coefficient maps A and B, which is where the analysis lives. The results below restate it for the system object of **??**: the variation-of-constants formula is a *trajectory* in the sense of **??**, which the stability theory quantifies over.

2.6.5 The time-invariant case

For a time-invariant system the state transition matrix is the matrix exponential, so the variation-of-constants formula collapses to the closed form every textbook writes.

2.7 Discrete-time systems

For $x(t+1) = A(t)x(t)$ the state transition matrix is an ordered product rather than a series, and the whole theory becomes finite: every proof below is an induction, with no convergence, no differentiation and no Gronwall estimate. Two differences from the continuous case are substantive rather than merely technical. There is no standing bound on A to carry, since a finite product always converges; and Φ need not be invertible, so the semigroup property genuinely requires $\tau \le s \le t$ instead of holding in any order.

Definition 57 (Discrete state transition matrix)**.** The discrete-time *state transition matrix* $\Phi(t, t_0)$:

$$\Phi(t, t_0) := \begin{cases} A(t-1)A(t-2)\cdots A(t_0+1)A(t_0) & t > t_0, \\ I & t \le t_0. \end{cases}$$

Formally, the ordered product of A over $[t_0, t)$, which for $t \le t_0$ is the empty product I. The system runs forward only, so the $t < t_0$ branch carries no meaning; it is there to make the definition total.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, equation (5.11).

Lemma 58 (The discrete state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Property P5.5.*

Proof. The defining product is over the empty range $[t_0, t_0)$. □

Lemma 59 (One-step recursion)**.** *For $t \ge t_0$, $\Phi(t+1, t_0) = A(t)\,\Phi(t, t_0)$. With lemma 58 this is the existence half of Hespanha's P5.5: the discrete state transition matrix solves the matrix recursion.*

Proof. Peel the leading factor $A(t)$ off the defining product. □

Theorem 60 (Discrete state transition matrix: uniqueness)**.** *Hespanha, P5.5. If $Z(t_0) = I$ and $Z(t+1) = A(t)Z(t)$ for all $t \ge t_0$, then $Z(t) = \Phi(t, t_0)$ for all $t \ge t_0$.*

Proof. Induction on t from t_0: the base case is lemma 58 and the step chains the recursion against lemma 59. No analytic content at all, unlike the continuous-time counterpart — the recursion pins each value outright. □

Theorem 61 (Uniqueness of the discrete state response)**.** *If $z(t_0) = x_0$ and $z(t+1) = A(t)z(t)$ for all $t \ge t_0$, then $z(t) = \Phi(t, t_0)x_0$ for all $t \ge t_0$: the vector form of theorem 60.*

Proof. □

Theorem 62 (Columns of the discrete state transition matrix)**.** *Hespanha, P5.6. For each i, the i-th column of $\Phi(t, t_0)$ is the unique solution of $z(t+1) = A(t)z(t)$ with $z(t_0) = e_i$, for $t \geq t_0$.*

Proof. theorem 61 at $x_0 = e_i$, using $\Phi(t, t_0)e_i = (\Phi(t, t_0))_{\cdot,i}$. □

Theorem 63 (Semigroup property, discrete)**.** *Hespanha, P5.7. For $\tau \leq s \leq t$,*

$$\Phi(t, s)\,\Phi(s, \tau) = \Phi(t, \tau).$$

Unlike the continuous-time case there is no inverse: Φ is a product of matrices that need not be nonsingular, so the ordering $\tau \leq s \leq t$ is a genuine restriction here rather than an artifact.

Proof. Induction on t from s, chaining lemma 59 at both s and τ. □

2.7.1 Discrete variation of constants

Lemma 64 (Discrete variation of constants at the initial time)**.** *The discrete variation-of-constants formula*

$$x(t) := \Phi(t, t_0)x_0 + \sum_{\tau=t_0}^{t-1} \Phi(t, \tau+1)B(\tau)u(\tau)$$

takes the value x_0 at $t = t_0$: the forcing sum is over the empty range and $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Section 5.3 (discrete variation of constants).*

Proof. □

Theorem 65 (Discrete variation of constants: existence)**.** *For $t \geq t_0$, the formula of lemma 64 satisfies the forced recursion*

$$x(t+1) = A(t)\,x(t) + B(t)\,u(t).$$

Proof. Peel the top term off the sum defining $x(t+1)$ and fold $A(t)\Phi(t, \tau+1)$ into $\Phi(t+1, \tau+1)$ termwise by lemma 59, applied once more to the leading $\Phi(t, t_0)$ term. The peeled term is exactly $B(t)u(t)$, since $\Phi(t+1, t+1) = I$. □

Theorem 66 (Discrete variation of constants: uniqueness)**.** *Any two solutions of $z(t+1) = A(t)z(t) + B(t)u(t)$ with the same value at t_0 agree for all $t \geq t_0$.*

Proof. Induction on t from t_0. Unlike theorem 56, no Lipschitz condition or Gronwall estimate is needed: the recursion determines each successive value outright, so uniqueness is immediate. □

2.7.2 The LTI case: the matrix exponential

For a constant state matrix the Peano–Baker series sums to the matrix exponential, which is Hespanha's (6.1)–(6.2). The point of proving it is economy: every property of Φ established above for a time-varying A then holds for $e^{A(t-t_0)}$ by specialization, with no separate LTI theory. The two exponential facts that need no solution theory at all — the semigroup law (P6.3) and the finite-polynomial form via Cayley–Hamilton (P6.5) — are proved as statements about exp and A alone.

Lemma 67 (Peano–Baker terms for a constant matrix)**.** *For a constant state matrix A, the k-th Peano–Baker term collapses to*

$$\Phi_k(t, t_0) = \frac{(t - t_0)^k}{k!} A^k.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.1).*

Proof. Induction on k. The $k = 0$ term is the identity. For the step, $\Phi_{k+1}(t, t_0) = \int_{t_0}^{t} A\,\Phi_k(s, t_0)\,ds$; substituting the inductive hypothesis pulls A^{k+1} out of the integral and leaves $\int_{t_0}^{t}(s - t_0)^k ds = (t - t_0)^{k+1}/(k + 1)$, which combines with $1/k!$ to give $1/(k + 1)!$. □

Lemma 68 (The LTI state transition matrix as a series)**.** *For a constant state matrix A, the state transition matrix is the power series*

$$\Phi(t, t_0) = \sum_{k=0}^{\infty} \frac{(t - t_0)^k}{k!} A^k.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.1).*

Proof. Sum lemma 67 termwise; the Peano–Baker series converges for any continuous state matrix, a constant one included. □

Theorem 69 (The LTI state transition matrix is $e^{A(t-t_0)}$)**.** *For a constant state matrix A, the state transition matrix is the matrix exponential:*

$$\Phi(t, t_0) = e^{A(t-t_0)}.$$

This is the bridge that lets every property already proved for Φ in the time-varying setting — existence, uniqueness, the semigroup law, invertibility, variation of constants — be reused for LTI systems by specialization rather than reproved.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.2).*

Proof. Both sides are power series in A: lemma 68 for the left, the defining series of exp for the right. Match them termwise. □

Theorem 70 (Uniqueness of the LTI state response)**.** *Let A be constant and let z be a continuous integral solution of $\dot{x} = Ax$, $x(t_0) = x_0$, on $[t_0, t_1]$. Then $z(t) = e^{A(t-t_0)}x_0$ for every t in that interval.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, Property P6.1.*

Proof. The time-varying uniqueness theorem applied at the constant state matrix $A(\cdot) \equiv A$, whose bound $\|A(t)\| \le \|A\|$ is immediate, then rewritten through theorem 69. There is no new ODE content: uniqueness for LTI is uniqueness for LTV at a constant matrix. □

Theorem 71 (Columns of the LTI state transition matrix)**.** *Let A be constant. For each i, the i-th column of $e^{A(t-t_0)}$ is the unique continuous integral solution of $\dot{x} = Ax$ with $x(t_0) = e_i$, the i-th standard basis vector. Equivalently, the columns of the matrix exponential are the solutions from the standard basis.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, Property P6.2.*

Proof. theorem 70 at $x_0 = e_i$, in the column form supplied by the time-varying column-uniqueness theorem. □

2.7.3 Complexification

Spectral arguments must happen over $\mathbb{C}$, because $\mathbb{R}$ is not algebraically closed, while the systems of interest have real coefficients. Entrywise complexification is the bridge, and these record what it commutes with. The one with content is the transfer of positive definiteness: a real positive-definite matrix remains positive definite against *complex* vectors, which is what lets a Lyapunov certificate be evaluated against an eigenvector.

2.7.4 The matrix exponential as an algebraic object

These say nothing about trajectories, initial times or state transition matrices — they are facts about exp and A alone, and are proved as such. The complexification bridge is what lets spectral arguments, which must happen over $\mathbb{C}$, yield bounds on a real exponential; P6.5 collapses the exponential series to a finite polynomial in A. The eigenpair lemma is the half of spectral mapping that is pure power-series algebra, and is what turns an eigenvalue condition on A into a statement about the growth of e^{At}.

Lemma 72 (Complexification commutes with the matrix exponential)**.** *Entrywise complexification commutes with the matrix exponential: for a real matrix A,*

$$\overline{\exp(A)} = \exp(\overline{A}),$$

where $\overline{\ }$ denotes applying $\mathbb{R} \hookrightarrow \mathbb{C}$ entrywise. This is the bridge used to transfer spectral facts, which live over $\mathbb{C}$, to norm bounds on a real exponential.

Proof. Entrywise complexification is a continuous ring homomorphism, and exp commutes with any such. □

Theorem 73 (Semigroup property of the matrix exponential)**.** *Hespanha, P6.3. For every $t, \tau \in \mathbb{R}$,*

$$e^{At} e^{A\tau} = e^{A(t+\tau)}.$$

Proof. tA and τA commute, being scalar multiples of the same matrix, so the exponential of the sum factors. The general $e^X e^Y = e^{X+Y}$ needs commutativity and is false without it; here it is free. □

Theorem 74 (The matrix exponential as a finite polynomial)**.** *Hespanha, P6.5. There are scalar functions $\alpha_0, \dots, \alpha_{n-1} : \mathbb{R} \to \mathbb{R}$ with*

$$e^{At} = \sum_{i=0}^{n-1} \alpha_i(t)\, A^i \qquad \text{for every } t.$$

The matrix exponential — an infinite series in A — is a finite *polynomial in A of degree less than n, at the cost of coefficients that are transcendental in t.*

Proof. Cayley–Hamilton reduces every power A^k to a combination of $A^0, \dots, A^{n-1}$, say with coefficients $\bar{a}_i(k)$; substituting into the exponential series and exchanging the finite sum over i with the infinite sum over k exhibits $\alpha_i(t) = \sum_k t^k \bar{a}_i(k)/k!$ as a witness.

The exchange needs those series to converge, which holds because the reduced coefficients grow at most geometrically, $|\bar{a}_i(k)| \le CM^k$, dominating each α_i by a scalar exponential series. That

growth bound is the crux of the proof and is established by iteration, not by spectral theory: "multiply by X, reduce modulo the characteristic polynomial" is a fixed linear operation on the n-dimensional space of polynomials of degree $< n$, so one step scales the ℓ^1 size of the coefficient vector by at most a constant factor. No eigenvalues and no Jordan form appear anywhere. $\square$

Chapter 3

Ordinary differential equations

3.1 Continuity toolbox

Two facts about continuous real functions and curves, with no differential-equation content of their own, used throughout the chapters that follow: the comparison lemma needs the last root before a sign change, and the Lyapunov stability arguments need the first time a curve reaches a sphere.

Lemma 75 (Greatest zero of a sign-changing continuous function)**.** *Let g be continuous on $[t_0, t_1]$ with $g(t_0) \le 0 < g(t_1)$. Then g has a* last *root: there is $a \in [t_0, t_1)$ with $g(a) = 0$ and $g(t) > 0$ for every $t \in (a, t_1]$.*

Proof. □

Theorem 76 (First crossing of a sphere)**.** *Let φ be continuous on $[t_0, T]$ with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < \rho$, and suppose some $t_1 \in [t_0, T]$ satisfies $\rho \le \|\varphi(t_1) - x_{\mathrm{eq}}\|$. Then there is a* first *hitting time $\tau \in [t_0, T]$ of the sphere of radius ρ:*

$$\|\varphi(\tau) - x_{\mathrm{eq}}\| = \rho, \qquad \|\varphi(s) - x_{\mathrm{eq}}\| \le \rho \quad \forall\, s \in [t_0, \tau].$$

Proof. Take the minimum of the closed set $[t_0, T] \cap \|\varphi(\cdot) - x_{\mathrm{eq}}\|^{-1}\{\rho\}$, which is nonempty by the intermediate value theorem. Minimality rules out an earlier excursion outside the ball: one would force a second crossing before τ, again by the intermediate value theorem. □

3.2 Interval integrals

Estimates and integrability side conditions for the interval integral, used wherever a differential equation is handled in integral form. The last three concern an integral whose endpoints move: normalizing by the window length makes it a mean value, after which continuity, antitonicity and decay each need only a hypothesis on the endpoints themselves. They support the class $\mathcal{KL}$ construction in the last chapter.

Lemma 77 (Difference of integrals over adjacent intervals)**.** *For μ continuous on $[a, t]$ and $s \in [a, t]$,*

$$\int_a^t \mu - \int_a^s \mu = \int_s^t \mu.$$

Proof. □

Lemma 78 (Integral bound from a pointwise multiplicative bound)**.** *Let u take values in a normed space and v be real, both integrable on $[a,b]$ with $a \le b$. If $\|u(s)\| \le L\, v(s)$ for all $s \in [a,b]$, then*

$$\Big\| \int_a^b u \Big\| \le L \int_a^b v.$$

Proof. □

Lemma 79 (Integral bound from a constant pointwise bound)**.** *If $\|u(s)\| \le C$ for all $s \in [a,b]$ with $a \le b$, then*

$$\Big\| \int_a^b u \Big\| \le C\,(b-a).$$

Proof. □

Lemma 80 (Integral of a constant)**.** $\int_a^b C \,\mathrm{d}s = (b-a)\,C$ *for a real constant C.*

Proof. □

Lemma 81 (Interval integrability of a continuous composition)**.** *If f is jointly continuous and z is continuous on the segment between t_0 and t_1, then $s \mapsto f(s, z(s))$ is interval-integrable between them. This is the integrability side condition every integral-form solution argument needs.*

Proof. □

Lemma 82 (Differentiating an integral in its upper limit)**.** *For μ continuous on $[a,b]$ and $t \in (a,b)$,*

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big|_{x=t} \int_a^x \mu(\tau)\,\mathrm{d}\tau \;=\; \mu(t).$$

Proof. □

Lemma 83 (Continuity of an integral in its upper limit)**.** *If f is integrable on $[a,t]$ with $a \le t$, then $s \mapsto \int_a^s f(\tau)\,\mathrm{d}\tau$ is continuous on $[a,t]$.*

Proof. □

Lemma 84 (Continuity of an integral in both endpoints)**.** *Let f be locally integrable on $(0,\infty)$, and let a, b be continuous on a set S with $a(\eta), b(\eta) > 0$ in S. Then*

$$\eta \;\longmapsto\; \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is continuous on S.

Proof. □

Lemma 85 (The sliding average of an antitone function is antitone)**.** *Let f be antitone and locally integrable on $(0,\infty)$, and let a, b be monotone on a set S with $0 < a(\eta) < b(\eta)$. Then the mean value*

$$\eta \;\longmapsto\; \frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is antitone on S.

Proof. □

Lemma 86 (Sliding average of an eventually vanishing function)**.** *Let f be nonneg, antitone and locally integrable on $(0,\infty)$ with $f(s) \to 0$ as $s \to +\infty$, and let $a(\eta) < b(\eta)$ with $a(\eta) \to +\infty$ as $\eta \to +\infty$. Then*

$$\frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x \;\longrightarrow\; 0 \qquad (\eta \to +\infty).$$

The mean is squeezed between 0 and $f(a(\eta))$; no hypothesis relating a to b is needed.

Proof. □

3.3 Limits superior

Real-valued lim sup is an infimum of upper bounds, so it is only well-behaved where the function is bounded. The facts collected here are stated over an arbitrary filter and have no differential-equation content; the Dini derivative of the next section is the only current consumer.

Lemma 87 (Limsup is unchanged by adding a null term)**.** *Let l be a nontrivial filter, let f be bounded above and below along l, and let $g \to 0$ along l. Then* $\limsup_l (f+g) = \limsup_l f$.

Proof. One inequality is subadditivity of lim sup together with $\limsup_l g = 0$. For the other, write $f = (f+g) + (-g)$ and apply the same argument, $-g$ also tending to 0; the boundedness of $f+g$ needed for this follows from that of f and g separately. □

3.4 Dini derivatives

The upper right Dini derivative replaces the ordinary derivative where a function is only known to be continuous, which is the situation the comparison lemma is in. It is defined here as a lim sup valued in $\mathbb{R}$ rather than in $[-\infty,+\infty]$, so most results below carry a hypothesis that the forward difference quotients are bounded: without it an unbounded quotient makes the lim sup an infimum over the empty set, which $\mathbb{R}$ reports as 0.

Definition 88 (Upper right Dini derivative)**.** The *upper right Dini derivative* of $f : \mathbb{R} \to \mathbb{R}$ at t is

$$D^+ f(t) \;=\; \limsup_{h \to 0^+} \frac{f(t+h)-f(t)}{h}.$$

When f is differentiable at t this equals $f'(t)$.

3.4.1 Relationship to the classical derivative

Lemma 89 (Forward difference quotients of a differentiable function)**.** *If f has right derivative L at t, then $\frac{f(t+h)-f(t)}{h} \to L$ as $h \to 0^+$.*

Proof. □

Lemma 90 (Dini derivative of a differentiable function)**.** *If f has right derivative L at t, then $D^+f(t) = L$: the Dini derivative agrees with the classical one wherever the latter exists.*

Proof. □

3.4.2 Characterization by difference quotients

Lemma 91 (Upper bound on the Dini derivative)**.** *Let the forward difference quotients of f at t be bounded above and cobounded below along $h \to 0^+$. Then $D^+f(t) \le L$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ is eventually at most $L + \varepsilon$ as $h \to 0^+$.*

Proof. □

Lemma 92 (Lower bound on the Dini derivative)**.** *Under the same boundedness conditions as lemma 91, $L \le D^+f(t)$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ exceeds $L - \varepsilon$ frequently as $h \to 0^+$.*

Proof. □

3.4.3 Linearity properties

Lemma 93 (Subadditivity of the Dini derivative)**.** *The upper right Dini derivative is subadditive: $D^+(f+g)(t) \le D^+f(t) + D^+g(t)$, provided the forward difference quotients of f are bounded above and below and those of g are bounded above and cobounded below.*

Proof. □

Lemma 94 (Adding a differentiable function)**.** *If the forward difference quotients of f at t are bounded above and below, and g is differentiable at t, then $D^+(f+g)(t) = D^+f(t) + g'(t)$. Subadditivity becomes equality once one summand has a genuine derivative.*

Proof. □

Lemma 95 (Adding a linear function)**.** *If the forward difference quotients of f at t are bounded above and below, then $D^+(f + c\,\mathrm{id})(t) = D^+f(t) + c$ — the case $g(s) = cs$ of lemma 94, and the form the comparison arguments use.*

Proof. □

Lemma 96 (Positive scalar multiple)**.** *For $c > 0$, and forward difference quotients of f at t bounded above and cobounded below, $D^+(cf)(t) = c\,D^+f(t)$. Positivity is essential: a negative constant reverses the order and turns the* lim sup *into a* lim inf.

Proof. Multiplication by $c > 0$ is an order isomorphism of $\mathbb{R}$, and a lim sup commutes with one. Going through the order isomorphism rather than through the defining infimum avoids the degenerate empty-inf case. □

Lemma 97 (Negation and the lower Dini derivative)**.** *If the forward difference quotients of f at t are bounded above and below, then $-D^+(-f)(t) \le D^+f(t)$. The left-hand side is the* lower *right Dini derivative, so this is the expected inequality $D_+f(t) \le D^+f(t)$ written in terms of D^+ alone.*

Proof. □

3.4.4 The comparison step

Lemma 98 (Dini derivative from a one-sided bound)**.** *Let $a < b$, let $z(a) = v(a)$ and $z(t) < v(t)$ for all $t \in (a, b]$, let z have right derivative d_z at a, and let the forward difference quotients of v at a be bounded above. Then $d_z \le D^+v(a)$.*

Proof. On $(0, b-a)$ the hypotheses give $z(a+h) - z(a) \le v(a+h) - v(a)$, so the forward quotient of z is eventually below that of v. The quotient of z converges to d_z, so d_z = lim sup of it, and lim sup is monotone. □

3.5 Gronwall–Bellman inequality

A function obeying an integral inequality is bounded by the solution of the corresponding integral equation. The two specializations are what the rest of the development actually calls: constant Λ gives exponential growth, and constant Λ and μ gives the $Ce^{\mu(t-a)}$ bound behind every continuous-dependence estimate.

Theorem 99 (Gronwall–Bellman inequality)**.** *Khalil, Lemma A.1. Let $\Lambda, \mu : [a, b] \to \mathbb{R}$ be continuous with $\mu \ge 0$, and let $y : [a, b] \to \mathbb{R}$ be continuous satisfying*

$$y(t) \;\le\; \Lambda(t) + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a, b].$$

Then

$$y(t) \;\le\; \Lambda(t) + \int_a^t \Lambda(s)\, \mu(s)\, e^{\int_s^t \mu(\tau)\, \mathrm{d}\tau}\, \mathrm{d}s \qquad \forall\, t \in [a, b].$$

Proof. An integrating-factor argument. Write $z(t) = \int_a^t \mu y$, $M(t) = \int_a^t \mu$, and $v = z + \Lambda - y$, so that $v \ge 0$ is exactly the hypothesis. The transform $w = e^{-M}z$ has, by the product rule and lemma 82,

$$\dot{w} = e^{-M}\mu\,(y - z) = e^{-M}\mu\,(\Lambda - v) \;\le\; e^{-M}\mu\Lambda,$$

the inequality because e^{-M}, μ and v are all nonnegative — this is the only place $\mu \ge 0$ is used, and it is essential: a sign change would reverse it. Integrating from a, where $w(a) = 0$, gives $w(t) \le \int_a^t e^{-M(s)}\mu(s)\Lambda(s)\, ds$. Multiplying by $e^{M(t)}$ and folding $e^{M(t)}e^{-M(s)} = e^{\int_s^t \mu}$ turns this into the stated bound on $z(t)$, and $y \le \Lambda + z$ finishes. □

Lemma 100 (Gronwall–Bellman with a constant rate)**.** *Khalil, Lemma A.1 (the case $\Lambda(t) \equiv \Lambda$ constant). Let μ be continuous and nonnegative on $[a, b]$, and let y be continuous with*

$$y(t) \;\le\; C + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a, b].$$

Then $y(t) \le C\, e^{\int_a^t \mu(\tau)\, \mathrm{d}\tau}$ on $[a, b]$.

Proof. theorem 99 at $\Lambda \equiv C$ gives $y(t) \le C + \int_a^t C\mu(s) e^{\int_s^t \mu}\, ds$, and the integral evaluates in closed form: $s \mapsto -C\, e^{M(t)-M(s)}$ is an antiderivative of the integrand, so the integral is $Ce^{M(t)} - C$. The two C's cancel. □

Lemma 101 (Gronwall–Bellman with constant data)**.** *Khalil, Lemma A.1 (the case Λ and $\mu \ge 0$ both constant). Let $\mu \ge 0$ be constant and let y be continuous with*

$$y(t) \;\le\; C + \int_a^t \mu\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a, b].$$

Then $y(t) \le C\, e^{\mu(t-a)}$ on $[a, b]$. This is the form the continuous-dependence estimates use, and the reason their bounds carry a factor $e^{L(t-t_0)}$.

Proof. lemma 100 with $\int_a^t \mu = \mu(t - a)$. □

3.6 Continuous dependence on parameters

Definition 102 (Integral solution)**.** A function x, defined on the segment between t_0 and t_1, is an *integral solution* of $\dot{x} = F(t, x)$ with initial value x_0 when

$$x(t) \;=\; x_0 + \int_{t_0}^t F(s,\, x(s))\, \mathrm{d}s$$

for every t between t_0 and t_1.

Lemma 103 (Re-anchoring an integral solution)**.** *Let x be a continuous integral solution of $\dot{x} = F(t, x)$ with initial value x_0 on the segment between t_0 and t_1, with F jointly continuous. Then for any s, r on that segment, x is an integral solution on the segment between s and r, with initial value $x(s)$:*

$$x(t) \;=\; x(s) + \int_s^t F(w, x(w))\, \mathrm{d}w \qquad \textit{for } t \textit{ between } s \textit{ and } r.$$

Leaving both endpoints free means the domain need not shrink — taking $s = t_1$ and $r = t_0$ re-anchors at one end of the original segment while retaining all of it.

Proof. Subtract the defining equations at t and at s and split the integral at s, which is legitimate because $w \mapsto F(w, x(w))$ is interval-integrable on each piece. □

Lemma 104 (Time reflection of an integral solution)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ with initial value x_0 on the segment between t_0 and t_1. Then $\sigma \mapsto x(t_0 + t_1 - \sigma)$ is an integral solution, with the same initial value x_0, of the reflected field*

$$(r, y) \;\longmapsto\; -F(t_0 + t_1 - r,\; y)$$

on the segment between t_1 and t_0.

The map $\sigma \mapsto t_0 + t_1 - \sigma$ is an involution exchanging the two endpoints, so reflection turns a statement anchored at t_0 into one anchored at t_1 and nothing more. Its purpose is to let an argument that is inherently forward-marching — a Grönwall bootstrap, say — be applied unchanged in backward time.

Proof. Substituting $r \mapsto t_0 + t_1 - r$ in the defining integral reverses the orientation of the interval, contributing one sign, and the negation of the field contributes another; the two cancel. □

3.6.1 The integral and differential formulations

An ODE can be written as a differential equation, $\dot{x} = F(t, x)$, or as the equivalent integral equation. The continuous-dependence and Gronwall arguments want the integral form, which presupposes no differentiability and carries its initial condition; the Lyapunov arguments want the differential form, so that $V \circ \varphi$ can be differentiated. The results below are the two directions of the fundamental theorem of calculus relating them.

They are provision rather than current practice: nothing in the stability chapters has yet been routed through this equivalence, so it is the intended path from a trajectory to the results of this chapter rather than an exercised one.

Lemma 105 (Integral curves are integral solutions)**.** *Let x be an integral curve of F on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral solution on that segment, anchored at its own value $x(t_0)$:*

$$x(t) \;=\; x(t_0) + \int_{t_0}^{t} F(s, x(s))\,\mathrm{d}s.$$

Continuity of x itself is not assumed — it follows from differentiability.

Proof. The fundamental theorem of calculus. The derivative hypothesis holds on an open neighbourhood of each interior point, so it yields the right-derivative form the theorem needs, and continuity of F along x gives the integrability side condition. □

Lemma 106 (Integral solutions are integral curves)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral curve of F on that segment: it is differentiable, with $\dot{x}(t) = F(t, x(t))$.*

Proof. Differentiating the primitive: the integral term has derivative $F(t, x(t))$ by the fundamental theorem of calculus, and the constant x_0 contributes nothing. Transport that along the defining equation, which says x agrees with the primitive. □

Lemma 107 (Integral and differential forms agree)**.** *Given continuity of $s \mapsto F(s, x(s))$ on the segment between t_0 and t_1, the two formulations agree: x is an integral solution anchored at $x(t_0)$ if and only if it is an integral curve of F there. Continuity of F along x is exactly the price of the equivalence — without it the integral form is the weaker notion.*

Proof. The two implications are lemma 106 and lemma 105. □

Lemma 108 (Integral and differential forms agree, forward time)**.** *lemma 107 in forward time: for $t_0 \le t_1$, the equivalence holds over $[t_0, t_1]$. This is the interval the finite-forward stability predicates are phrased over, so this is the form the stability track would use to reach the results of this chapter.*

Proof. □

Lemma 109 (Integral form for an autonomous field)**.** *Let g be a continuous* autonomous *field and let x be an integral curve of g on $[t_0, t_1]$, $t_0 \le t_1$. Then x is an integral solution there, anchored at $x(t_0)$.*

This is the autonomous, forward-time specialization of lemma 108, and so the shape a trajectory of an autonomous system has: continuity of the field is the only hypothesis, the continuity of x along which F must be continuous being supplied by the curve itself.

Proof. □

3.6.2 Time invariance and local existence

An autonomous field has no preferred time origin, which is what lets the autonomous stability predicates fix the initial time at 0.

Lemma 110 (Time invariance)**.** *If x is an integral curve of an autonomous field g on a set s, then $t \mapsto x(t + \Delta t)$ is an integral curve of the same g on $s - \Delta t$, for every Δt.*

An autonomous field has no preferred time origin, so translating a solution translates its interval of definition and changes nothing else. For a time-varying field the translated solution would solve the translated *equation instead.*

Proof. □

Lemma 111 (Shifting a solution segment to the origin)**.** *lemma 110 in the form the finite-segment predicates need: a solution segment of an autonomous field on $[t_0, t_1]$ becomes one on $[0, t_1 - t_0]$ under $s \mapsto x(s + t_0)$.*

This is why the autonomous stability predicates may fix the initial time at 0 without loss: any segment can be shifted there.

Proof. □

Lemma 112 (Local existence for a C^1 field)**.** *Let g be an autonomous field that is C^1 at x_0. Then there is a $T > 0$ and a curve φ with $\varphi(0) = x_0$ that is an integral curve of g on $[0, T]$: a nontrivial solution segment exists from every point at which the field is C^1.*

The anchor 0 is a construction choice rather than a restriction — by lemma 110 a segment from any other anchor follows.

Reference: the Picard–Lindelöf local existence theorem.

Proof. Picard–Lindelöf local existence: being C^1 at x_0 gives a closed ball on which g is Lipschitz, and hence a solution on some $(-\varepsilon, \varepsilon)$; take $T := \varepsilon/2$. □

Theorem 113 (Picard–Lindelöf on a compact interval)**.** *Let $f : \mathbb{R} \to E \to E$ be jointly continuous and K-Lipschitz in the state variable, uniformly in time, on a Banach space E. Then for every $t_0 \le t_1$ and every $x_0 \in E$ there is an α with $\alpha(t_0) = x_0$ solving $\dot\alpha = f(t, \alpha)$ on all of $[t_0, t_1]$.*

Global — rather than merely local — Lipschitz continuity is what makes the solution reach t_1: it forces linear growth in the state, ruling out finite-time blowup.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2 (Global Existence and Uniqueness), proved in his Appendix C.1. Difference: Khalil assumes f piecewise continuous in t; here it is jointly continuous.*

Proof. Mathlib's Picard–Lindelöf theorem supplies a solution on a time interval of length Δt only when $\|f\| \le L$ on a ball of radius a about x_0 with $L\,\Delta t \le a$. For a K-Lipschitz field the sharpest available bound on that ball is $L = M + Ka$, where M bounds $\|f(t, x_0)\|$, so the condition becomes $M\Delta t \le a(1 - K\Delta t)$ — solvable in a exactly when $K\Delta t < 1$. One application therefore never covers a long interval.

Take instead a step h with $Kh \le 1/2$; then $a = 2Mh$ and $L = 2M$ satisfy the hypotheses, since $\|f(t, x)\| \le M + Ka = M(1 + 2Kh) \le 2M$ and $Lh = 2Mh = a$. This gives a solution on $[s, s + h]$ from any initial state. Chain n such steps, splicing consecutive segments at their shared endpoint: away from a junction the glued curve agrees with one piece on a neighbourhood, and at a junction the two one-sided derivatives combine. Choosing n with $nh \ge t_1 - t_0$ covers $[t_0, t_1]$. □

Lemma 114 (Right derivatives of an integral curve)**.** *If α is an integral curve of f on $[t_0, t_1]$, then for every $s \in [t_0, t_1)$ it has right derivative $f(s, \alpha(s))$ at s, i.e. α has derivative $f(s, \alpha(s))$ within $[s, \infty)$ at s.*

Proof. □

Lemma 115 (Gronwall separation of two solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$. Then*

$$d(\alpha(t), \beta(t)) \le d(\alpha(t_0), \beta(t_0))\, e^{K(t - t_0)} \qquad \forall\, t \in [t_0, t_1].$$

Uniqueness (lemma 116) is the case where the initial distance is zero; continuous dependence on the initial state is the statement read as a bound on the whole interval.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.4, in the unperturbed case: two solutions of the* same *field, so the perturbation bound is zero.*

Proof. □

Lemma 116 (Uniqueness of solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$ with $\alpha(t_0) = \beta(t_0)$. Then $\alpha = \beta$ on $[t_0, t_1]$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), the uniqueness half of Theorem 3.2.*

Proof. The initial distance in lemma 115 is zero, so the bound forces the distance to vanish throughout. □

Theorem 117 (Picard–Lindelöf on the forward ray)**.** *Let f be jointly continuous and K-Lipschitz in the state variable, uniformly in time. Then for every t_0 and x_0 there is an α with $\alpha(t_0) = x_0$ solving $\dot\alpha = f(t, \alpha)$ on all of $[t_0, \infty)$, and any other such solution agrees with it on $[t_0, \infty)$.*

Uniqueness is agreement on $[t_0, \infty)$, not equality of functions: the defining condition says nothing about $t < t_0$, so solutions may differ there.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, whose t_1 is arbitrary, extended to the forward ray. Uniqueness is stated as agreement on $[t_0, \infty)$, not as equality of functions.*

Proof. For each n take a solution α_n on $[t_0, t_0+n]$ (theorem 113). By uniqueness (lemma 116) these agree wherever two of them are both defined, so $\Phi(t) := \alpha_{\lceil t-t_0 \rceil}(t)$ is well defined and agrees with α_n on all of $[t_0, t_0+n]$. Near any $t \geq t_0$ choose $n > t - t_0$; then $[t_0, t_0+n]$ is a neighbourhood of t within $[t_0, \infty)$ on which $\Phi = \alpha_n$, so Φ inherits the derivative there. Uniqueness on the ray follows by applying the compact-interval uniqueness at each n. □

Theorem 118 (Picard–Lindelöf, scalar integral form)**.** *Let $g : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ be jointly continuous and L-Lipschitz in its state argument, uniformly in time, and let $t_0 \leq t_1$. Then for every x_0 there is a z, continuous on $[t_0, t_1]$, that is an integral solution of $\dot{z} = g(t, z)$ with $z(t_0) = x_0$ and has right derivative $g(s, z(s))$ at each $s \in [t_0, t_1)$.*

This is the scalar specialization of theorem 113; it is the existence hypothesis that theorem 122 takes as an assumption rather than discharging.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, scalar case.*

Proof. Apply theorem 113 with $E = \mathbb{R}$. Continuity on $[t_0, t_1]$ is automatic for an integral curve, which converts the differential form into the integral one (lemma 108); the right derivatives are lemma 114. □

3.6.3 Continuous dependence

Two solutions of the same equation, started at different points or subject to different perturbations, separate no faster than exponentially — at a rate fixed by the Lipschitz constant of the field. Everything downstream that needs uniqueness gets it from here, by taking the two solutions to agree initially and the perturbation to vanish, which collapses the bound to zero.

Theorem 119 (Continuous dependence on initial states and parameters)**.** *Khalil, Theorem 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on the segment between t_0 and t_1, let y be an integral solution of $\dot{y} = f(t, y)$ with $y(t_0) = y_0$, and let z be an integral solution of the perturbed equation $\dot{z} = f(t, z) + g(t, z)$ with $z(t_0) = z_0$, where $\|g(t, x)\| \leq \mu$. Then for every t on that segment,*

$$\|y(t) - z(t)\| \;\leq\; \|y_0 - z_0\|\, e^{L|t-t_0|} \;+\; \frac{\mu}{L}\big(e^{L|t-t_0|} - 1\big).$$

Two solutions may separate, but no faster than exponentially, at a rate set by the Lipschitz constant alone; the two error sources — a different starting point and a perturbed field — contribute additively. The bound is stated on the unordered segment, so it runs in either time direction.

The perturbation g is assumed only bounded and integrable along z, never continuous, so z need not be differentiable anywhere. That is why the theorem is phrased over the integral formulation rather than the differential one.

Proof. The forward case is a Grönwall bootstrap. Subtracting the two integral equations and applying the triangle inequality gives

$$\|y(\tau) - z(\tau)\| \;\leq\; \|y_0 - z_0\| + \mu(\tau - t_0) + L \int_{t_0}^{\tau} \|y(s) - z(s)\|\, \mathrm{d}s,$$

the Lipschitz hypothesis bounding the f-difference and the uniform bound on g the rest. Adding μ/L to both sides absorbs the linear term $\mu(\tau - t_0)$ into the integral, putting the inequality in the shape lemma 101 expects, and Grönwall delivers the exponential.

That bootstrap is inherently forward-marching, so rather than repeat it for the backward case, reflect: lemma 104 sends $\sigma \mapsto t_0 + t_1 - \sigma$ and negates the field, turning the backward instance into a forward one on $[t_1, t_0]$ to which the same argument applies unchanged. □

Theorem 120 (Continuous dependence on parameters)**.** *Khalil, Theorem 3.5. If y solves $\dot{y} = f(t, y)$ and z solves $\dot{z} = f(t, z)+g(t, z)$ with $\|g(t, x)\| \le \alpha$ and $\|z_0-y_0\| \le \alpha$, and $\alpha(1+1/L)e^{L(t_1-t_0)} \le \varepsilon$, then $\|y(t) - z(t)\| \le \varepsilon$ for all $t \in [t_0, t_1]$.*

Proof. Reduce to theorem 99 applied to $\|y-z\|$, using the L-Lipschitz bound on f and the α-bound on g. □

3.7 Comparison lemma

The comparison lemma bounds a function satisfying a differential *inequality* by the solution of the corresponding differential *equation*. The inequality is stated with the Dini derivative, so v need only be continuous — which is what makes the lemma applicable to $\|x(t)\|$, not differentiable where a trajectory crosses the origin.

Lemma 121 (Integral form from a pointwise derivative)**.** *Let u be continuous on $[t_0, t_1]$ with $u(t_0) = u_0$ and $\dot{u}(t) = f(t, u(t))$ at every interior point, and let f be jointly continuous. Then u is an integral solution:*

$$u(s) = u_0 + \int_{t_0}^{s} f(\tau, u(\tau))\,\mathrm{d}\tau.$$

.

Proof. The fundamental theorem of calculus on $[t_0, s]$; the integrand is continuous there, being a composition of continuous maps. □

Theorem 122 (Comparison lemma)**.** *Khalil, Lemma 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on $[t_0, t_1]$, and suppose*

1. *u is continuous on $[t_0, t_1]$ and solves $\dot{u} = f(t, u)$ with $u(t_0) = u_0$;*
2. *v is continuous on $[t_0, t_1]$ with $v(t_0) \le u_0$, satisfies the Dini subsolution inequality $D^+v(t) \le f(t, v(t))$, and has forward difference quotients bounded above at each t;*
3. *for every $\lambda > 0$ the perturbed equation $\dot{z} = f(t, z) + \lambda$, $z(t_0) = u_0$, has a solution on $[t_0, t_1]$.*

Then $v(t) \le u(t)$ for all $t \in [t_0, t_1]$.

The perturbed solutions in (3) are not constructed here — the caller supplies them. They do exist under the hypotheses already listed, since $f + \lambda$ is Lipschitz exactly when f is, so Picard–Lindelöf applies on $[t_0, t_1]$; but that existence step is an axiom in this development, and leaving it as a hypothesis is what keeps this theorem, and everything downstream of it, free of any axiom beyond Lean's own.

Proof. Fix $\lambda > 0$ and let z_λ be as in hypothesis (3).

Claim: $v \le z_\lambda$ *on* $[t_0, t_1]$. Otherwise $v(t_{\text{bad}}) > z_\lambda(t_{\text{bad}})$ for some t_{bad}, and since $v(t_0) \le z_\lambda(t_0)$ the continuous difference $v - z_\lambda$ has a last zero $a < t_{\text{bad}}$, with $v > z_\lambda$ on $(a, t_{\text{bad}}]$. Being below v immediately to the right of a forces $\dot{z}_\lambda(a) \le D^+v(a)$, so

$$f(a, z_\lambda(a)) + \lambda \;\le\; D^+v(a) \;\le\; f(a, v(a)) = f(a, z_\lambda(a)),$$

contradicting $\lambda > 0$.

Now let $\varepsilon > 0$ and choose λ small enough that theorem 120 — which applies because f is Lipschitz in its state argument, and which treats λ as a perturbation of size λ — gives $\|u - z_\lambda\| \le \varepsilon/2$ on $[t_0, t_1]$. Then $v \le z_\lambda \le u + \varepsilon$. As ε was arbitrary, $v \le u$. □

Chapter 4

Lyapunov stability

4.1 Trajectories, equilibria, and comparison functions

Definition 123 (Solution segment)**.** A *solution segment* of $\dot{x} = f(x)$ on $[t_0, t_1]$ is an integral curve of the vector field restricted to that interval. Unlike a globally defined trajectory it need not exist for all time, so quantifying over segments does not silently discard solutions with a finite escape time.

Definition 124 (Equilibrium point)**.** A point $x_{\mathrm{eq}} \in \mathbb{R}^n$ is an *equilibrium* of $\dot{x} = f(x)$ when $f(x_{\mathrm{eq}}) = 0$.

Definition 125 (Class $\mathcal{K}$ function)**.** A *class* $\mathcal{K}$ function on $[0, a)$ is a continuous strictly increasing map $\alpha : [0, a) \to [0, b)$ with $\alpha(0) = 0$. All class K functions have an inverse $\alpha^{-1} : [0, b) \to [0, a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}$). Differences:

- Khalil writes the codomain as $[0, \infty)$; here the bound b is tracked in the type.
- The inverse is bundled as data rather than derived.

Definition 126 (Class $\mathcal{K}_\infty$ function)**.** A *class* $\mathcal{K}_\infty$ function is a continuous strictly increasing map $\alpha : [0, \infty) \to [0, \infty)$ with $\alpha(0) = 0$ and $\alpha(r) \to \infty$ as $r \to \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}_\infty$). The inverse is bundled as data rather than derived.

4.1.1 Building class $\mathcal{K}$ functions

A recognition criterion and the closure properties. The restatements of the definition itself (strict monotonicity as an iff, the two inverse equations, and so on) are left to the Lean API; they say nothing the definition above does not.

Definition 127 (Class $\mathcal{K}$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$ and $f(a) = b$ and be continuous and strictly increasing on $[0, a]$, with $a, b > 0$. Then f is a class $\mathcal{K}$.

Definition 128 (Restriction of a class $\mathcal{K}$ function)**.** If α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and $0 < c < a$, then α restricted to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, \alpha(c))$.

Definition 129 (Restriction to a prescribed codomain)**.** definition 128 with the codomain bound named: if $\alpha(c) = e$, the restriction of α to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, e)$.

Definition 130 (Inverse of a class $\mathcal{K}$ function)**.** Class $\mathcal{K}$ is closed under inversion. The inverse of a class K function is also class K.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, first bullet. Khalil's domain $[0, \alpha(a))$ is the codomain bound b carried by the type.

Definition 131 (Composition of class $\mathcal{K}$ functions)**.** Class $\mathcal{K}$ is closed under composition: if α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and β is class $\mathcal{K}$ on $[0, b) \to [0, c)$, then $\beta \circ \alpha$ is class $\mathcal{K}$ on $[0, a) \to [0, c)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, third bullet.

4.1.2 Building class $\mathcal{K}_\infty$ functions

The same recognition criterion and closure properties on the unbounded domain, together with the closure properties that need the global domain (sum, positive scaling, minimum), the power functions as the standard examples, and the restriction back to class $\mathcal{K}$.

Definition 132 (Class $\mathcal{K}_\infty$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$, be continuous and strictly increasing on $[0, \infty)$, and have $f(r) \to \infty$ as $r \to \infty$. Then f is a class $\mathcal{K}_\infty$ function, its inverse being the inverse of f on $[0, \infty)$.

Definition 133 (Inverse of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under inversion: if α is class $\mathcal{K}_\infty$, so is α^{-1}.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, second bullet.

Definition 134 (Composition of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under composition: if α and β are class $\mathcal{K}_\infty$, so is $\beta \circ \alpha$, with inverse $\alpha^{-1} \circ \beta^{-1}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fourth bullet.

Definition 135 (Sum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise addition: if α and β are class $\mathcal{K}_\infty$, so is $\alpha + \beta$.

Definition 136 (Positive multiple of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under multiplication by a positive constant: if α is class $\mathcal{K}_\infty$ and $c > 0$, so is $c\alpha$.

Definition 137 (Minimum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise minimum: if α and β are class $\mathcal{K}_\infty$, so is $\min(\alpha, \beta)$.

Definition 138 (Powers of a class $\mathcal{K}_\infty$ function)**.** For every $p > 0$, the map $r \mapsto r^p$ is class $\mathcal{K}_\infty$.

Definition 139 (A class $\mathcal{K}_\infty$ function restricts to class $\mathcal{K}$)**.** Every class $\mathcal{K}_\infty$ function α restricts, for each $a > 0$, to a class $\mathcal{K}$ function on $[0, a) \to [0, \alpha(a))$.

Lemma 140 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let $\omega : \mathbb{R} \to \mathbb{R}$ satisfy $\omega(0) = 0$ and be monotone on $[0, \infty)$. Then there is a class $\mathcal{K}_\infty$ function α with $\omega(r) \le \alpha(r)$ for all $r \ge 0$.*

Proof. The smoothing axiom supplies a continuous, strictly increasing f with $f(0) = 0$, $f(r) \to \infty$ and $\omega \le f$ on $[0, \infty)$; by definition 132 that f is class $\mathcal{K}_\infty$. □

4.2 Stability predicates

The predicates below quantify over every solution segment on a compact interval $[t_0, t_1]$. This prevents finite escape from making a stability claim vacuous merely because a non-equilibrium solution is not globally defined. A solution segment is an integral curve of the vector field restricted to that interval, in the sense of `IsTrajectoryOn`.

Definition 141 (Lyapunov stability)**.** An equilibrium is Lyapunov stable when every finite forward solution segment starting sufficiently close remains within any prescribed neighborhood for its entire interval of definition.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.1 (*stable*), for $\dot{x} = f(x)$. Differences:

- Khalil fixes $t_0 = 0$; here it is arbitrary. Immaterial for an autonomous field.
- Khalil presumes solutions on $[0, \infty)$; here the quantification is over finite forward segments, so the predicate is not vacuous when solutions escape in finite time.

Definition 142 (Local exponential stability)**.** An equilibrium is locally exponentially stable on finite forward segments when nearby solutions satisfy a uniform estimate $\|x(t) - x_{\mathrm{eq}}\| \le C e^{-a(t-t_0)} \|x(t_0) - x_{\mathrm{eq}}\|$.

Reference: Khalil, *Nonlinear Systems*.

Definition 143 (Local asymptotic stability)**.** An equilibrium is *locally asymptotically stable* when it is forward Lyapunov stable and there is a radius $c > 0$ such that every solution defined for all forward time with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < c$ satisfies $\varphi(t) \to x_{\mathrm{eq}}$ as $t \to \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.1 (*asymptotically stable*), for $\dot{x} = f(x)$. Differences:

- Khalil reuses the stability δ as the attractivity radius; here c is separate.
- Convergence is asserted only of forward-complete solutions.

Definition 144 (Global asymptotic stability)**.** An equilibrium is *globally asymptotically stable* when it is forward Lyapunov stable and every solution defined for all forward time satisfies $\varphi(t) \to x_{\mathrm{eq}}$ as $t \to \infty$.

Reference: Khalil, *Nonlinear Systems*.

Definition 145 (Instability)**.** Instability is the negation of stability quantified over all finite forward solution segments.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.1 (*unstable*), for $\dot{x} = f(x)$.

4.3 Sublevel sets and positive invariance

Definition 146 (Sublevel set)**.** The *sublevel set* of $V : \mathbb{R}^n \to \mathbb{R}$ at level $c \in \mathbb{R}$ is

$$\Omega_c(V) \;=\; \{\, x \in \mathbb{R}^n \;:\; V(x) \le c \,\}.$$

Definition 147 (Positively invariant set)**.** A set $S \subseteq \mathbb{R}^n$ is *positively invariant* for $\dot{x} = f(x)$ when every solution segment φ on $[t_0, t_1]$ starting in S remains in S throughout:

$$\varphi(t_0) \in S \;\Rightarrow\; \varphi(t) \in S \quad \forall\, t \in [t_0, t_1].$$

Lemma 148 (Sublevel sets of a radially unbounded function are compact)**.** *If $V : \mathbb{R}^n \to \mathbb{R}$ is continuous and radially unbounded ($V(x) \to \infty$ as $\|x\| \to \infty$), then every sublevel set $\Omega_c(V)$ (definition 146) is compact.*

Proof. Closed: $\Omega_c(V) = V^{-1}((-\infty, c])$ by continuity. Bounded: coercivity yields R with $\Omega_c(V) \subseteq \overline{B}(0, R)$. Compact: Heine–Borel in $\mathbb{R}^n$. □

Lemma 149 (Forward invariance of sublevel sets)**.** *Let V be a local Lyapunov certificate on D and let the sublevel set $\{V \le c\}$ be contained in D. If $V(\varphi(t_0)) < c$ then $V(\varphi(t)) < c$ for every t in the segment: a solution starting strictly inside a sublevel set contained in the certificate domain never leaves it.*

Proof. Suppose not, and let T^* be the first time at which $c \le V(\varphi(t))$; the set of such times is closed, so the infimum is attained, and $T^* > 0$ because $V(\varphi(t_0)) < c$. Before T^* the solution stays in $\{V < c\} \subseteq D$, so lemma 160 applies on $[t_0, T^*]$ and gives $V(\varphi(T^*)) \le V(\varphi(t_0)) < c$, contradicting $c \le V(\varphi(T^*))$. □

4.4 Lyapunov functions

Definition 150 (Local Lyapunov function)**.** A function $V : \mathbb{R}^n \to \mathbb{R}$ is a *local Lyapunov function* on an open domain $D \ni x_{\text{eq}}$ when V is continuous and differentiable on all of $\mathbb{R}^n$, $V(x_{\text{eq}}) = 0$, $V > 0$ on $D \setminus \{x_{\text{eq}}\}$, and the Lie derivative satisfies $\dot{V}(x) = \nabla V(x) \cdot f(x) \le 0$ for all $x \in D$.

Global regularity of V, with positivity and the Lie-derivative sign required only on D, is what lets the chain-rule and intermediate-value arguments be applied uniformly. Note this notion does *not* require x_{eq} to be an equilibrium; the strict variants below do.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.1, hypotheses (4.2) and (4.3), for $\dot{x} = f(x)$.

Definition 151 (Strict local Lyapunov function)**.** A *strict local Lyapunov function* on D strengthens definition 150 in three ways: V is required to be C^1 rather than merely differentiable; x_{eq} must be an equilibrium, $f(x_{\text{eq}}) = 0$; and the Lie derivative is strictly negative, $\dot{V}(x) < 0$ for all $x \in D \setminus \{x_{\text{eq}}\}$. It additionally requires a $c > 0$ with $\Omega_c(V) \subseteq D$ compact (definition 146), which replaces radial unboundedness in the local setting.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.1, hypotheses (4.2) and (4.4), for $\dot{x} = f(x)$. The equilibrium and compact-sublevel-set fields are additional: Khalil obtains the compact set inside the proof, and leaves $f(x_{\text{eq}}) = 0$ to the standing setup of (4.1).

Definition 152 (Strict Lyapunov function)**.** A *global strict Lyapunov function* for $\dot{x} = f(x)$ at x_{eq} is a C^1 map $V : \mathbb{R}^n \to \mathbb{R}$ with $f(x_{\text{eq}}) = 0$, $V(x_{\text{eq}}) = 0$, $V > 0$ everywhere else, $\dot{V}(x) < 0$ on $\mathbb{R}^n \setminus \{x_{\text{eq}}\}$, and all sublevel sets $\Omega_c(V)$ compact (coercivity).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.2, hypotheses (4.5) and (4.7), for $\dot{x} = f(x)$. Differences:

- Radial unboundedness (4.6) is replaced by compactness of every sublevel set, which for continuous V it implies (lemma 148).
- The equilibrium field $f(x_{\text{eq}}) = 0$ is additional.

Definition 153 (Asymptotic Lyapunov function)**.** The classical GAS Lyapunov certificate: a C^1 map $V : \mathbb{R}^n \to \mathbb{R}$ with $f(x_{\mathrm{eq}}) = 0$, $V(x_{\mathrm{eq}}) = 0$, $V > 0$ elsewhere, $\dot{V} < 0$ on $\mathbb{R}^n \setminus \{x_{\mathrm{eq}}\}$, and radially unbounded ($V(x) \to \infty$ as $\|x\| \to \infty$). Implies definition 152 via lemma 148.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.2, hypotheses (4.5), (4.6) and (4.7), for $\dot{x} = f(x)$. The equilibrium field $f(x_{\mathrm{eq}}) = 0$ is additional, left by Khalil to the standing setup of (4.1).

4.4.1 The hierarchy of certificates

The four notions above differ only in how much they demand; each stronger one weakens to the next, so a theorem proved for the weakest applies to all of them.

Lemma 154 (An asymptotic Lyapunov function is strict)**.** *An asymptotic Lyapunov function is a strict one: radial unboundedness supplies the bounded-sublevel-set condition the strict notion requires.*

Proof. A continuous radially unbounded function has compact sublevel sets. □

Lemma 155 (A strict Lyapunov function is a Lyapunov function)**.** *A strict Lyapunov function is a local Lyapunov function on all of $\mathbb{R}^n$: strict negativity away from the equilibrium weakens to nonpositivity everywhere.*

Proof. Away from x_{eq} the strict bound gives the weak one. At x_{eq} itself, $f(x_{\mathrm{eq}}) = 0$ and a continuous linear map sends 0 to 0, so the Lie derivative vanishes. □

Lemma 156 (A strict local Lyapunov function is a local one)**.** *A strict* local *Lyapunov function on D is a local Lyapunov function on the same D — lemma 155 without the global hypotheses.*

Proof. □

4.5 Differentiating along solutions

Every Lyapunov argument begins the same way: compose the certificate with a solution and differentiate. The chain rule gives the Lie derivative $DV(\varphi(t))[f(\varphi(t))]$, and a sign condition on it becomes a monotonicity statement about $V \circ \varphi$. Collecting that step here fixes one spelling of the Lie derivative for the whole development.

Lemma 157 (Chain rule along a trajectory)**.** *Let V be differentiable and let φ be a solution segment on $[t_0, t_1]$. Then at every interior time t,*

$$(V \circ \varphi)'(t) \;=\; DV(\varphi(t))\,[\,f(\varphi(t))\,].$$

Every Lyapunov argument in the library opens with this composition; stating it once fixes the spelling of the Lie derivative at every call site.

Proof. □

Lemma 158 (Rate-bounded monotonicity)**.** *Let φ be an integral curve of f on a convex set s, with $DV(\varphi(t))[f(\varphi(t))] \le -c$ at every interior point. Then $t \mapsto V(\varphi(t)) + ct$ is antitone on s.*

The weight is what turns a bound on the rate *into a bound on* elapsed time*: since V is bounded below, antitonicity of $V \circ \varphi + ct$ caps how long φ can remain where the rate bound holds. Taking $c = 0$ gives plain monotonicity.*

Proof. Differentiate: by the chain rule along the solution the derivative is $DV(\varphi(t))[f(\varphi(t))] + c \le 0$ by the rate bound, and a function with nonpositive derivative on a convex set is antitone there. □

Lemma 159 (V is nonincreasing along a solution)**.** *The unweighted case $c = 0$ of lemma 158: $V \circ \varphi$ is antitone wherever the Lie derivative is nonpositive. This is the sense in which a Lyapunov function "never increases along solutions".*

Proof. □

Lemma 160 (V does not increase from the initial time)**.** *lemma 159 in endpoint form: if V is a local Lyapunov certificate on D and the solution segment stays in D throughout $[t_0, t_1]$, then $V(\varphi(t_1)) \le V(\varphi(t_0))$.*

Proof. □

4.6 Milestone theorems for autonomous systems

Theorem 161 (Lyapunov's stability theorem)**.** *If V is a local Lyapunov function (definition 150) for $\dot{x} = f(x)$ on a domain $D \ni x_{\mathrm{eq}}$, then x_{eq} is stable on every finite forward solution segment (definition 141).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.1, first conclusion ($x = 0$ is stable under (4.2) and (4.3)).*

Proof. Pick ε_0 so $\overline{B}(x_{\mathrm{eq}}, \varepsilon_0) \subseteq D$. Let $m = \min_{S_{\varepsilon'}} V > 0$. Choose δ with $V < m$ on $B(x_{\mathrm{eq}}, \delta)$. If $\|\varphi(t^*) - x_{\mathrm{eq}}\| \ge \varepsilon$, monotonicity of V gives $V(\varphi(t^*)) \le V(\varphi(0)) < m \le V(\varphi(t^*))$, a contradiction. □

Theorem 162 (Lyapunov's local asymptotic stability theorem)**.** *If V is a strict local Lyapunov function (definition 151) and f is continuous, then x_{eq} is locally asymptotically stable (definition 143).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.1, second conclusion (strengthening (4.3) to (4.4) gives asymptotic stability).*

Proof. The compact sublevel set $\Omega_{c_0} \subseteq D$ is positively invariant, so the Lie derivative is at most $-\gamma < 0$ off any ball around x_{eq}, draining V at a definite rate; the δ-ball supplied by stability is therefore reached in finite time, and stability applied at the entry time confines the solution thereafter. □

Theorem 163 (Lyapunov's global asymptotic stability theorem)**.** *If V is a global strict Lyapunov function (definition 152) and f is continuous, then x_{eq} is globally asymptotically stable (definition 144).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.2, with radial unboundedness (4.6) in its compact-sublevel-set form.*

Proof. Stability from theorem 161 fixes δ for the given ε. On the compact set $\{V \le M\} \cap \{\delta \le \|x - x_{\mathrm{eq}}\|\}$ the Lie derivative is at most $-\gamma < 0$, so V is drained at a definite rate and the δ-ball is reached in finite time; stability applied at the entry time keeps the solution within ε thereafter. □

Theorem 164 (Global asymptotic stability from a radially unbounded V)**.** *If V is a radially unbounded strict Lyapunov function (definition 153) and f is continuous, then x_{eq} is globally asymptotically stable on finite forward solution segments.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.2 (Barbashin–Krasovskii).*

Proof. Radial unboundedness gives compact sublevel sets (lemma 148), so V satisfies definition 152; apply theorem 163. □

The asymptotic results rest on a uniformity that mere per-solution convergence does not give: outside a ball about the equilibrium the Lie derivative is bounded away from zero, so every solution is drained at a shared rate.

Lemma 165 (Uniform entry time)**.** *Let V be a local Lyapunov certificate on D with $DV(x)[f(x)] < 0$ away from the equilibrium, and let $\{V \le M\}$ be a compact subset of D. For every $\delta > 0$ there is a $\tau \ge 0$ such that any solution segment starting in $\{V \le M\}$, staying in D, and remaining at distance at least δ from x_{eq} throughout, has length at most τ.*

The point is that τ depends only on M and δ, not on the segment. This uniformity is what separates asymptotic stability from mere per-solution convergence.

Proof. On the compact set $\{V \le M\} \cap \{\delta \le \|x - x_{\mathrm{eq}}\|\}$ the Lie derivative is continuous and strictly negative, hence bounded above by some $-c < 0$. By lemma 158, $V(\varphi(t)) + ct$ is antitone along the segment, and V is bounded below on the sublevel set, which caps the elapsed time. □

4.6.1 Stability from exponential decay, and instability from escape

Theorem 166 (Exponential stability implies Lyapunov stability)**.** *Local exponential stability implies Lyapunov stability.*

Reference: Khalil, Nonlinear Systems*.*

Proof. Given ε, take $\delta := \min(r, \varepsilon/C)$. The decay estimate $\|\varphi(t) - x_{\mathrm{eq}}\| \le Ce^{-a(t-t_0)}\|\varphi(t_0) - x_{\mathrm{eq}}\|$ and $e^{-a(t-t_0)} \le 1$ for $t \ge t_0$ give $\|\varphi(t) - x_{\mathrm{eq}}\| < \varepsilon$. □

Theorem 167 (Instability from a fixed escape radius)**.** *Fix $\varepsilon > 0$. If for every $\delta > 0$ some solution segment starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable. This is the contrapositive of stability, in the form the instability theorems produce their witnesses.*

Reference: Khalil, Nonlinear Systems*.*

Proof. □

4.7 Lyapunov's indirect method

Definition 168 (Affine-linear vector field)**.** For a matrix $A \in \mathbb{R}^{n\times n}$ and a point $x_{\mathrm{eq}} \in \mathbb{R}^n$, the *affine-linear vector field* is

$$f(x) = A\,(x - x_{\mathrm{eq}}),$$

for which x_{eq} is an equilibrium.

Lemma 169 (Absorbing the linearization remainder)**.** *Let f be C^1 with $Df(x_{\mathrm{eq}}) = A$, let M be a matrix, and let $c > 0$. Then there is $r > 0$ such that for every x with $\|x - x_{\mathrm{eq}}\| < r$,*

$$\left| D\, q_M(x)\, [\, f(x) - f(x_{\mathrm{eq}}) - A(x - x_{\mathrm{eq}})\,] \right| \le c\, \|x - x_{\mathrm{eq}}\|^2,$$

where q_M is the centred quadratic form $q_M(x) = \langle M(x - x_{\mathrm{eq}}),\, x - x_{\mathrm{eq}} \rangle$.

The bracket is the first-order remainder of f at x_{eq}. The point is that c is arbitrary*: near enough to the equilibrium the remainder perturbs $\dot{q}_M$ by less than any prescribed multiple of $\|x - x_{\mathrm{eq}}\|^2$, which is the order of $\dot{q}_M$ itself. This is why the linearization decides stability.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7 (the quadratic-Lyapunov proof of Lyapunov's indirect method).*

Proof. Write $y = x - x_{\mathrm{eq}}$ and e for the remainder. Differentiability of f at x_{eq} gives, for any $\eta > 0$, a radius within which $\|e\| \le \eta \|y\|$. The derivative of a centred quadratic form is bounded by $|Dq_M(x)[e]| \le 2\|M\|\, \|y\|\, \|e\|$, so the product is at most $2\|M\|\eta\, \|y\|^2$. Choosing $\eta = c/(4(\|M\| + 1))$ makes $2\|M\|\eta \le c$. □

Theorem 170 (Instability of an affine-linear system)**.** *If $A \in \mathbb{R}^{n \times n}$ has a complex eigenvalue μ with $\operatorname{Re}\mu > 0$, then x_{eq} is an unstable equilibrium (definition 145) of the affine-linear field $x \mapsto A(x - x_{\mathrm{eq}})$ (definition 168).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7.*

Proof. Let $v \neq 0$ be an eigenvector for μ and consider the real eigenmode $t \mapsto x_{\mathrm{eq}} + \rho \operatorname{Re}(e^{\mu t} v)$, a genuine solution of the linear equation for every $\rho > 0$. Its initial displacement is $\rho\|v\|$, which can be made smaller than any δ, while some coordinate has modulus growing like $\rho\, |v_i|\, e^{(\operatorname{Re}\mu) t}$ — the real part recovers at least half the complex amplitude over each quarter-period. Since $\operatorname{Re}\mu > 0$ this exceeds the fixed escape radius at some finite time, so no δ confines every solution segment. □

Theorem 171 (Quantitative linearization error bound)**.** *Let A be the Fréchet derivative of f at x_0. For every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$\|x - x_0\| < \delta \implies \|f(x) - f(x_0) - A(x - x_0)\| \le \varepsilon\, \|x - x_0\|.$$

This is the ε–δ form of $f(x) - f(x_0) - A(x - x_0) = o(\|x - x_0\|)$.

Proof. □

The stable-branch theorem gives uniform local exponential decay on every finite forward solution segment, without silently excluding solutions that are not globally defined.

Theorem 172 (Lyapunov's indirect method, stable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1, and let x_{eq} be an equilibrium. If its Jacobian A at the equilibrium is Hurwitz, then there are uniform local constants giving exponential decay along every finite forward solution segment that starts sufficiently close to x_{eq}.*

Reference: Khalil, Nonlinear Systems*.*

Proof. Solve the identity-forced Lyapunov equation for a positive-definite P, use $V(x) = (x - x_{\mathrm{eq}})^{\mathsf{T}} P (x - x_{\mathrm{eq}})$, absorb the $o(\|x - x_{\mathrm{eq}}\|)$ linearization remainder on a small ball, and apply a weighted-energy estimate up to the first possible exit time. □

Theorem 173 (Exponential Chetaev instability criterion)**.** *Suppose the vector field f and certificate V are C^1, V is quadratically bounded on a ball, its positive set accumulates at the base point, and its Lie derivative satisfies $\dot{V} \geq 2\alpha V$ there for some $\alpha > 0$. Then the base point is unstable with respect to finite forward solution segments.*

Reference: Hahn, Stability of Motion *(Chetaev's instability method).*

Proof. Globalize the vector field by a smooth cutoff, integrate the differential inequality on a sufficiently long finite segment, and stop the curve at its first crossing of the certificate ball. □

Theorem 174 (Lyapunov's indirect method, unstable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1 with $f(x_{\mathrm{eq}}) = 0$. If the Jacobian at x_{eq} has a complex eigenvalue with positive real part, then x_{eq} is unstable when stability is quantified over every finite forward solution segment.*

Reference: Khalil, Nonlinear Systems *(Lyapunov's indirect method).*

Proof. Build the real quadratic certificate supplied by theorem 33, absorb the first-order nonlinear remainder on a small ball, and apply the exponential Chetaev criterion. □

4.8 LaSalle's invariance principle

Lyapunov's theorems conclude stability from a strict decrease of V. LaSalle's principle covers the case $\dot{V} \leq 0$, where V may stall: a trajectory confined to a compact set must then settle into the region where $\dot{V}$ vanishes. The mechanism is the ω-limit set. Along a trajectory V is nonincreasing and bounded below, hence convergent, so V is *constant* on $\omega(\varphi)$ — and a set on which a solution can live while V stays constant is one where $\dot{V} = 0$. Making that last step a theorem rather than a picture is what the invariance of $\omega(\varphi)$ supplies, and it is the only place uniqueness of solutions is needed.

Definition 175 (ω-limit set of a trajectory)**.** The *ω-limit set* of a trajectory φ is

$$\omega(\varphi) = \bigcap_{T \geq 0} \overline{\{\varphi(t) : t \geq T\}},$$

the set of points y such that $\varphi(t)$ is frequently in every neighbourhood of y as $t \to \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), §4.2, where it is written L^+ and called the *positive limit set*.

Lemma 176 (Membership in the ω-limit set)**.** *$y \in \omega(\varphi)$ (definition 175) if and only if for every neighbourhood U of y there are arbitrarily large t with $\varphi(t) \in U$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), §4.2. Khalil characterizes L^+ by convergent sequences $x(t_i) \to y$ with $t_i \to \infty$; the frequently-in-every-neighbourhood form above is the same condition stated with filters.*

Proof. □

Lemma 177 (The ω-limit set is positively invariant)**.** *Let f be Lipschitz and let φ be a solution of $\dot{x} = f(x)$ on $[0, \infty)$. Then the ω-limit set $\omega(\varphi)$ (definition 175) is positively invariant (definition 147): every solution segment starting in $\omega(\varphi)$ remains in $\omega(\varphi)$.*

Neither boundedness of φ nor compactness of an enclosing set is needed; those hypotheses of Khalil's Lemma 4.1 are what make $\omega(\varphi)$ nonempty and compact, not what make it invariant.

Proof. Let ψ solve the equation on $[t_0, t_1]$ with $p = \psi(t_0) \in \omega(\varphi)$, and fix $t \in [t_0, t_1]$. Given $\varepsilon > 0$, put $C = e^{K(t_1 - t_0)}$ and pick, by the definition of the ω-limit set, arbitrarily large r with $\|\varphi(r) - p\| < \varepsilon / C$. The time-shifted curve $u(\cdot) = \varphi(\cdot + r - t_0)$ solves the same autonomous equation on $[t_0, t_1]$ and satisfies $u(t_0) = \varphi(r)$, so Grönwall gives

$$\|\varphi(r + t - t_0) - \psi(t)\| = \|u(t) - \psi(t)\| \le \|\varphi(r) - p\|\, e^{K(t - t_0)} < \varepsilon.$$

Since r may be taken arbitrarily large, so may $r + t - t_0$; hence φ is frequently within ε of $\psi(t)$, i.e. $\psi(t) \in \omega(\varphi)$.

Khalil argues instead with the flow map and the semigroup identity $\phi(t + t_i; x_0) = \phi(t; \phi(t_i; x_0))$, which presumes a globally defined flow. Comparing the two solutions directly needs only uniqueness, in its quantitative Grönwall form. □

Definition 178 (The set where the Lie derivative vanishes)**.** For $\dot{x} = f(x)$, a function V and a set Ω, put

$$E = \{x \in \Omega : \dot{V}(x) = DV(x)\,[\,f(x)\,] = 0\},$$

the subset of Ω on which the Lie derivative of V along f vanishes.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.4, where it is the set E.

Lemma 179 (The Lie derivative vanishes on the ω-limit set)**.** *Let Ω be compact and positively invariant for $\dot{x} = f(x)$ with f of class C^1 and Lipschitz, let $V \in C^1$ satisfy $\dot{V} \le 0$ on Ω, and let φ be a solution on $[0, \infty)$ with $\varphi(0) \in \Omega$. Then*

$$\dot{V}(y) = DV(y)\,[\,f(y)\,] = 0 \qquad \text{for every } y \in \omega(\varphi).$$

Equivalently $\omega(\varphi) \subseteq E = \{x : \dot{V}(x) = 0\}$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), proof of Theorem 4.4.*

Proof. Along the trajectory V is nonincreasing and bounded below on the compact set Ω, so $V(\varphi(t)) \to L$ for some L; by continuity of V every ω-limit point sees that same value, so $V \equiv L$ on $\omega(\varphi)$. Fix $y \in \omega(\varphi)$ and let ψ solve $\dot{x} = f(x)$ on $[0, T]$, $T > 0$, with $\psi(0) = y$; such a ψ exists because f is C^1. By lemma 177 the whole segment $\psi([0, T])$ lies in $\omega(\varphi)$, so $V \circ \psi \equiv L$ there. A constant function has vanishing derivative, and the chain rule identifies that derivative at 0 with $DV(\psi(0))[\dot{\psi}(0)] = DV(y)[f(y)]$. Uniqueness of the derivative within $[0, T]$ at its left endpoint (valid because $T > 0$) gives $DV(y)[f(y)] = 0$. □

Theorem 180 (LaSalle's invariance principle)**.** *Let Ω be compact and positively invariant for $\dot{x} = f(x)$ with f of class C^1 and Lipschitz, let $V \in C^1$ satisfy $\dot{V}(x) \le 0$ on Ω, and put*

$$E = \{x \in \Omega : \dot{V}(x) = 0\}$$

(definition 178). Let M contain every positively invariant subset of E. Then every trajectory φ on $[0, \infty)$ starting in Ω satisfies $\varphi(t) \to M$ as $t \to \infty$.

Khalil takes M to be the largest invariant *subset of E. Here M must absorb every* positively *invariant subset, which is a larger family, so the hypothesis on M is correspondingly stronger; what is proved about $\omega(\varphi)$ is forward invariance (lemma 177), and backward invariance would require solutions through every ω-limit point to extend backwards in time.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.4.*

Proof. The ω-limit set lies in Ω, because Ω is closed and φ stays in it, and $\dot{V}$ vanishes on it (lemma 179); hence $\omega(\varphi) \subseteq E$. It is positively invariant by lemma 177, so the hypothesis on M gives $\omega(\varphi) \subseteq M$. Finally, a trajectory confined to a compact set converges to its own ω-limit set: for any open $W \supseteq M \supseteq \omega(\varphi)$, φ is eventually in W. □

Theorem 181 (Barbashin's theorem)**.** *Let $V \in C^1$ be a local Lyapunov function for $\dot{x} = f(x)$ on a domain D (definition 150). Suppose for some $c > 0$ the sublevel set $\Omega_c = \{V \leq c\}$ is contained in D, compact, and positively invariant, that f is C^1 and Lipschitz, and that $\{x_{\text{eq}}\}$ is the only positively invariant subset of $E = \{x \in \Omega_c : \dot{V}(x) = 0\}$ (definition 178). Then x_{eq} is locally asymptotically stable (definition 143).*

The hypothesis on E is the one a user of LaSalle actually discharges: solve $\dot{V} = 0$ and check that no solution can stay there but the equilibrium. Positive invariance of Ω_c is a hypothesis rather than a consequence of $\dot{V} \leq 0$: deducing it would need a solution through each point of Ω_c.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.1.*

Proof. □

Theorem 182 (Krasovskii's theorem)**.** *If $V \in C^1$ is positive definite, $\dot{V} \leq 0$ on $\mathbb{R}^n$ and radially unbounded, f is C^1 and Lipschitz, and for every c the only positively invariant subset of $E_c = \{x \in \Omega_c : \dot{V}(x) = 0\}$ (definition 178) is $\{x_{\text{eq}}\}$, then x_{eq} is globally asymptotically stable (definition 144).*

Unlike Barbashin's theorem (theorem 181) no positively invariant set is assumed: radial unboundedness makes every sublevel set compact, and $\dot{V} \leq 0$ then keeps the trajectory inside the one through its initial state.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.2.*

Proof. □

4.9 Lyapunov class $\mathcal{K}$ bounds

A Lyapunov function is positive definite: a pointwise condition, awkward to combine with estimates. The theorem below converts it into monotone bounds in the single variable $\|x\|$, after which every Lyapunov argument can be run in the comparison-function calculus. It rests on two smoothing results, which replace a monotone bound by a continuous strictly increasing one.

Lemma 183 (Class $\mathcal{K}$ minorant of a positive monotone function)**.** ***Assumed without proof.*** *Let $r > 0$ and let $\psi : \mathbb{R} \to \mathbb{R}$ satisfy $\psi(0) = 0$, $\psi(s) > 0$ for $s \in (0, r]$, and ψ nondecreasing on $[0, r]$. Then there are $b > 0$ and a function f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $f(s) \leq \psi(s)$ for all $s \in [0, r]$.*

That is: a positive nondecreasing function admits a class $\mathcal{K}$ minorant on any bounded interval. This is what turns a positive-definite V into the lower comparison bound of theorem 186.

Lemma 184 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** ***Assumed without proof.*** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ satisfy $\varphi(0) = 0$ and be monotone on $[0, \infty)$. Then there is an f with $f(0) = 0$, continuous and strictly increasing on $[0, \infty)$, $f(s) \to \infty$ as $s \to \infty$, and $\varphi(s) \leq f(s)$ for all $s \geq 0$.*

That is: a nondecreasing function admits a class $\mathcal{K}_\infty$ majorant. The unbounded domain is what distinguishes this from lemma 185, which it implies.

Lemma 185 (Class $\mathcal{K}$ majorant on a bounded interval)**.** *Let $r > 0$ and let φ satisfy $\varphi(0) = 0$ and be monotone on $[0, r]$. Then there are $b > 0$ and an f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $\varphi(s) \le f(s)$ for all $s \in [0, r]$.*

Proof. Clamp φ to $\varphi_{\text{ext}}(s) = \varphi(\min(s, r))$, which is monotone on all of $[0, \infty)$ and still vanishes at 0, and apply lemma 184. Restricting the resulting majorant to $[0, r]$ gives f; $b = f(r) > 0$ because f is strictly increasing from $f(0) = 0$. □

Theorem 186 (Class $\mathcal{K}$ sandwich bounds)**.** *Let $V : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\overline{B}(0, r)$ with $V(0) = 0$ and $V(x) > 0$ for every nonzero x in that ball. Then there are class $\mathcal{K}$ functions α_1, α_2 on $[0, r)$ with*

$$\alpha_1(\|x\|) \;\le\; V(x) \;\le\; \alpha_2(\|x\|) \qquad \forall\, \|x\| < r.$$

The conclusion holds on the open *ball: α_1 and α_2 are class $\mathcal{K}$ on $[0, r)$, so neither is defined at $\|x\| = r$.*

This is what lets a Lyapunov argument be run entirely in terms of comparison functions: positive definiteness is a pointwise condition, and this converts it into monotone bounds in the single variable $\|x\|$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.3 (proved in his Appendix C.4). Differences:*

- *The bound holds on the open ball $\|x\| < r$, with α_1, α_2 class $\mathcal{K}$ on $[0, r)$; Khalil states it on $[0, r]$.*
- *The radially unbounded $\mathcal{K}_\infty$ case is not derived here.*

Proof. For the lower bound take $\psi(s) = \inf\{V(x) : s \le \|x\| \le r\}$, the infimum of V over the annulus of inner radius s: it vanishes at 0, is positive for $s > 0$ by compactness and positive definiteness, and is nondecreasing. For the upper bound take $\phi(s) = \sup\{V(x) : \|x\| \le s\}$, similarly nondecreasing and vanishing at 0. Neither is continuous or strictly monotone, so smooth them: lemma 183 gives a class $\mathcal{K}$ minorant of ψ and lemma 185 a class $\mathcal{K}$ majorant of ϕ. Since $\psi(\|x\|) \le V(x) \le \phi(\|x\|)$ pointwise, these sandwich V. □

Chapter 5

Non-autonomous systems

For $\dot{x} = f(t, x)$ the initial time matters, and the definitions split accordingly: a bound may hold at each initial time with a margin depending on it, or with one margin serving every initial time. That distinction — plain versus *uniform* — is what the whole chapter turns on, and it has no counterpart in the autonomous theory, where time invariance lets the initial time be fixed at 0 without loss.

5.1 Trajectories and equilibria

Definition 187 (Trajectory of a time-varying system)**.** A *trajectory from* t_0 of the non-autonomous ODE $\dot{x} = f(t, x)$ is a map φ defined and satisfying $\dot{\varphi}(t) = f(t, \varphi(t))$ for every $t \geq t_0$. This is the object Khalil's definitions refer to as "the solution $x(t)$": forward-complete from its initial time, with existence assumed rather than proved.

Definition 188 (Equilibrium of a time-varying system)**.** A point $x_{\text{eq}} \in \mathbb{R}^n$ is an *equilibrium* of the non-autonomous ODE $\dot{x} = f(t, x)$ when $f(t, x_{\text{eq}}) = 0$ for every $t \in \mathbb{R}$.

5.2 Uniform convergence times and comparison-function bounds

The estimates the class $\mathcal{KL}$ characterization is assembled from. Naming them here, rather than re-typing their quantifiers at each use, is what makes the characterization statements readable.

Definition 189 (Local uniform convergence times)**.** The trajectories of $\dot{x} = f(t, x)$ starting within c of x_{eq} *have local uniform convergence times* when for every $\eta > 0$ there is $T(\eta) > 0$, independent of t_0 and of the solution, with

$$\|\varphi(t) - x_{\text{eq}}\| < \eta \qquad \forall\, t \geq t_0 + T(\eta),$$

for every $t_0 \geq 0$ and every trajectory φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\text{eq}}\| < c$.

Reference: This is Khalil (4.17), the attractivity half of uniform asymptotic stability.

Definition 190 (Global uniform convergence times)**.** The trajectories of $\dot{x} = f(t, x)$ *have global uniform convergence times* when they have local uniform convergence times (definition 189) from every radius $c > 0$. The delay may depend on the radius as well as the tolerance, $T = T(\eta, c)$.

Reference: This is Khalil (4.18).

Definition 191 (Uniform class $\mathcal{K}$ bound)**.** The trajectories of $\dot{x} = f(t, x)$ *have the uniform class* $\mathcal{K}$ *bound* α about x_{eq} when

$$\|\varphi(t) - x_{\text{eq}}\| \leq \alpha(\|\varphi(t_0) - x_{\text{eq}}\|)$$

for every $t_0 \geq 0$, every trajectory φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\text{eq}}\| < a$, and every $t \geq t_0$.

Reference: This is Khalil (4.19), the estimate that characterizes uniform stability.

Definition 192 (Uniform class $\mathcal{K}_\infty$ bound)**.** The trajectories of $\dot{x} = f(t, x)$ *have the uniform class* $\mathcal{K}_\infty$ *bound* α about x_{eq} when $\|\varphi(t) - x_{\text{eq}}\| \leq \alpha(\|\varphi(t_0) - x_{\text{eq}}\|)$ for every $t_0 \geq 0$, every trajectory φ on $[t_0, \infty)$, and every $t \geq t_0$, with no restriction on the initial state.

Reference: This is Khalil (4.20) in its class $\mathcal{K}\infty$ form, the estimate behind the *global* results.

5.3 Stability predicates

Definition 193 (Stability)**.** The equilibrium x_{eq} of $\dot{x} = f(t, x)$ is *stable* when

$$\forall \varepsilon > 0,\ \forall t_0 \geq 0,\ \exists\, \delta = \delta(\varepsilon, t_0) > 0,\ \forall \varphi \text{ on } [t_0, \infty),\ \|\varphi(t_0) - x_{\text{eq}}\| < \delta \ \Rightarrow\ \forall t \geq t_0,\ \|\varphi(t) - x_{\text{eq}}\| < \varepsilon.$$

Reference: This is Khalil (4.16).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*stable*), estimate (4.16). Khalil places the equilibrium at the origin; here it is a general x_{eq}.

Definition 194 (Uniform stability)**.** The equilibrium x_{eq} is *uniformly stable* when δ in definition 193 can be chosen independently of t_0:

$$\forall \varepsilon > 0,\ \exists\, \delta = \delta(\varepsilon) > 0,\ \forall t_0 \geq 0,\ \forall \varphi \text{ on } [t_0, \infty),\ \|\varphi(t_0) - x_{\text{eq}}\| < \delta \ \Rightarrow\ \forall t \geq t_0,\ \|\varphi(t) - x_{\text{eq}}\| < \varepsilon.$$

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly stable*): (4.16) with $\delta = \delta(\varepsilon)$ independent of t_0.

Definition 195 (Global uniform stability)**.** The equilibrium x_{eq} is *globally uniformly stable* when it is uniformly stable (definition 194) with a margin $\delta(\varepsilon)$ that can be chosen to satisfy $\lim_{\varepsilon \to \infty} \delta(\varepsilon) = \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4, the first clause of *globally uniformly asymptotically stable*: uniformly stable with $\delta(\varepsilon) \to \infty$.

Definition 196 (Instability, time-varying)**.** The equilibrium x_{eq} is *unstable* when it is not stable (definition 193).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*unstable*).

Lemma 197 (Uniform stability implies stability)**.** *Uniform stability (definition 194) implies stability (definition 193). The two differ only in the order of the quantifiers on δ and t_0, so the uniform margin serves at every initial time.*

Proof. □

Lemma 198 (Global uniform stability implies stability)**.** *Global uniform stability (definition 195) implies stability (definition 193).*

Proof. The margin $\delta(\varepsilon)$ is already independent of t_0, so it serves at every initial time. □

Lemma 199 (Trajectories from a stable equilibrium)**.** *Let x_{eq} be stable (definition 193) and let φ be a trajectory from t_0 with $\varphi(t_0) = x_{\mathrm{eq}}$. Then $\varphi(t) = x_{\mathrm{eq}}$ for every $t \geq t_0$: a solution sitting at a stable equilibrium cannot leave it.*

The hypothesis is stability, not $f(t, x_{\mathrm{eq}}) = 0$. "An equilibrium stays put" does not *follow from the vector field vanishing there — that also needs uniqueness of solutions. Without it a solution can leave: $\dot{x} = x^{2/3}$ has $f(0) = 0$, yet both $x \equiv 0$ and $x = (t/3)^3$ start at 0.*

Proof. Stability places $\varphi(t)$ within ε of x_{eq} for every $\varepsilon > 0$, so the distance is zero. □

5.4 Asymptotic and exponential stability

Definition 200 (Asymptotic stability)**.** The equilibrium x_{eq} is *asymptotically stable* when it is stable (definition 193) and for each $t_0 \geq 0$ there exists $c = c(t_0) > 0$ such that every solution defined for all forward time with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < c$ satisfies $\varphi(t) \to x_{\mathrm{eq}}$ as $t \to \infty$.

Attractivity quantifies over forward-complete solutions, as convergence must; the completeness restriction is harmless because the stability conjunct guards it.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*asymptotically stable*): stable, with an attractivity radius $c = c(t_0)$ that may depend on the initial time.

Definition 201 (Uniform asymptotic stability)**.** The equilibrium x_{eq} is *uniformly asymptotically stable* when it is uniformly stable (definition 194) and there exists $c > 0$ (independent of t_0) such that for each $\eta > 0$ there is $T = T(\eta) > 0$ with

$$\|\varphi(t) - x_{\mathrm{eq}}\| < \eta \quad \forall\, t \geq t_0 + T(\eta),\ \forall\, \|\varphi(t_0) - x_{\mathrm{eq}}\| < c,\ \forall\, t_0 \geq 0.$$

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly asymptotically stable*), estimate (4.17). Khalil writes the convergence-time clause inline; here it is factored out as definition 189.

Definition 202 (Global uniform asymptotic stability)**.** The equilibrium x_{eq} is *globally uniformly asymptotically stable* when it is uniformly stable with $\delta(\varepsilon) \to \infty$ as $\varepsilon \to \infty$, and for each pair of positive numbers η and c there is $T = T(\eta, c) > 0$ such that

$$\|\varphi(t) - x_{\mathrm{eq}}\| < \eta \quad \forall\, t \geq t_0 + T(\eta, c),\ \forall\, \|\varphi(t_0) - x_{\mathrm{eq}}\| < c,\ \forall\, t_0 \geq 0.$$

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*globally uniformly asymptotically stable*), estimate (4.18). The two clauses are factored as definition 195 and definition 190.

Definition 203 (Exponential stability)**.** The equilibrium x_{eq} is *exponentially stable* when there exist positive constants c, k, and λ such that

$$\|\varphi(t) - x_{\mathrm{eq}}\| \leq k\, \|\varphi(t_0) - x_{\mathrm{eq}}\|\, e^{-\lambda(t - t_0)}$$

for all $t \geq t_0 \geq 0$ and all trajectories φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < c$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5, estimate (4.21). The rate is written γ here and λ by Khalil.

Definition 204 (Global exponential stability)**.** The equilibrium x_{eq} is *globally exponentially stable* when the bound in definition 203 holds for every initial state $\varphi(t_0) \in \mathbb{R}^n$, i.e., $c = \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5, the global case: (4.21) holding for any initial state, i.e. with no radius c.

5.5 Uniform decay envelopes

A uniform convergence-time hypothesis says that for each tolerance η there is a delay $T(\eta)$ after which every trajectory is inside the η-ball. Read the other way round, it bounds how far a trajectory can still be after a given elapsed time — but only through a function of η that may be discontinuous, and so is not yet a comparison function.

The two results below do the regularization: they turn the delay into a genuine decay envelope, a function of elapsed time alone that is continuous, positive and vanishing at $+\infty$. That envelope is the time-decay half of the class $\mathcal{KL}$ bounds of the next section; the class $\mathcal{K}$ bound supplies the other half, the dependence on the initial deviation.

Lemma 205 (Local uniform decay envelope)**.** *Suppose the trajectories of $\dot{x} = f(t, x)$ have local uniform convergence times on the ball of radius c (definition 189) and admit a uniform class $\mathcal{K}$ bound α defined on $[0, a_\alpha)$ (definition 191). Let $0 < a \le c$ with $a < a_\alpha$. Then there is a singular class $\mathcal{L}$ function U (definition 221) such that*

$$\|\varphi(t) - x_{\mathrm{eq}}\| \le U(t - t_0)$$

for every $t_0 \ge 0$, every trajectory φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < a$, and every $t > t_0$.

The bound depends on the elapsed time alone: neither on t_0, nor on the trajectory, nor on where in the a-ball it started.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5.*

Proof. Let $\bar{T}(\eta, r)$ be the least delay after which every trajectory starting within r of x_{eq} lies inside the η-ball; it is finite by hypothesis and antitone in η, but possibly discontinuous. Smooth it by the sliding average

$$W(r, \eta) = \frac{2}{\eta} \int_{\eta/2}^{\eta} \left[\bar{T}(s, r) + \frac{r}{\eta} \right] ds,$$

which equals $\frac{2}{\eta} \int_{\eta/2}^{\eta} \bar{T}(s, r)\, ds + r/\eta$. The averaged term is antitone in η and the r/η term is strictly antitone, so $W(r, \cdot)$ is continuous and strictly antitone on $(0, \infty)$; it dominates r/η, hence tends to $+\infty$ as $\eta \to 0^+$, and $\bar{T}(\cdot, r)$ vanishes beyond $\alpha(r)$, hence $W(r, \eta) \to 0$ as $\eta \to \infty$. Inverting it (lemma 215, lemma 216, lemma 217, lemma 218, lemma 219) gives $U = \mathrm{invFunOn}\, W(r, \cdot)\, (0, \infty)$, singular class $\mathcal{L}$. Finally $W(r, U(s)) = s$ and $W \ge \bar{T} + r/\eta$ give $\bar{T}(U(s), r) < s$, so s is itself a valid convergence time to the $U(s)$-ball, which is the claimed bound. □

Lemma 206 (Global uniform decay envelope)**.** *Suppose the trajectories of $\dot{x} = f(t, x)$ have global uniform convergence times (definition 190) and admit a uniform class $\mathcal{K}_\infty$ bound (definition 192). Then there is a family $U : (0, \infty) \times (0, \infty) \to \mathbb{R}$ with*

1. *$U(r, s) > 0$ for all $r > 0$ and $s > 0$;*
2. *$U(r, \cdot)$ antitone on $(0, \infty)$ for each $r > 0$;*
3. *$U(r, s) \to 0$ as $s \to \infty$, for each $r > 0$;*
4. *$r \mapsto U(r + 1, s)$ monotone on $[0, \infty)$ for each $s > 0$;*
5. *$\|\varphi(t) - x_{\mathrm{eq}}\| \le U(r, t - t_0)$ for every $r > 0$, every $t_0 \ge 0$, every trajectory φ on $[t_0, \infty)$ with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < r$, and every $t > t_0$.*

Conditions (1)–(4) are exactly the hypotheses of lemma 236; (5) is the decay bound. Unlike the local case (lemma 205) no single envelope can serve every initial state, so the radius remains a parameter.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5 (global case).*

Proof. Every radius is finite, so at radius r the global hypotheses restrict to the local ones (uniform convergence times on the r-ball, and the class $\mathcal{K}_\infty$ bound read as a class $\mathcal{K}$ bound on $[0, r+1)$). Take $U(r,\cdot)$ to be the envelope of lemma 205 at that radius, namely the inverse of $W(r,\cdot)$. Conditions (1)–(3) and (5) are then that lemma's conclusions, and (4) holds because W is monotone in r — a larger ball has larger optimal convergence times — so inverting reverses nothing. □

5.6 Comparison-function characterizations

Each stability notion above is equivalent to the existence of a comparison-function bound. These equivalences are what let the Lyapunov theorems conclude a stability property by exhibiting a bound, and conversely let a stability hypothesis be used as an estimate. Both sides quantify over the same trajectories — those defined on $[t_0, \infty)$ — so each is a restatement of one class of solutions, not a relation between two.

Lemma 207 (Class $\mathcal{K}$ characterization of uniform stability)**.** *The equilibrium x_{eq} of $\dot{x} = f(t,x)$ is* uniformly stable *(definition 194) if and only if there exist a class $\mathcal{K}$ function α on $[0,a)$, with a independent of t_0, such that*

$$\|\varphi(t) - x_{\mathrm{eq}}\| \le \alpha(\|\varphi(t_0) - x_{\mathrm{eq}}\|) \quad \forall\, t \ge t_0 \ge 0,\ \forall\, \|\varphi(t_0) - x_{\mathrm{eq}}\| < a.$$

This is the first half of Khalil's Lemma 4.5; the displayed bound is his (4.19). Both sides quantify over the same trajectories — those defined on $[t_0,\infty)$ — so the equivalence is between two descriptions of one class of solutions, not between two classes.

Proof. □

Lemma 208 (Class $\mathcal{KL}$ characterization of uniform asymptotic stability)**.** *The equilibrium x_{eq} is* uniformly asymptotically stable *(definition 201) if and only if there exist a class $\mathcal{KL}$ function β and a positive constant c, independent of t_0, such that*

$$\|\varphi(t) - x_{\mathrm{eq}}\| \le \beta(\|\varphi(t_0) - x_{\mathrm{eq}}\|,\, t - t_0) \quad \forall\, t \ge t_0 \ge 0,\ \forall\, \|\varphi(t_0) - x_{\mathrm{eq}}\| < c.$$

This is the second half of Khalil's Lemma 4.5.

Proof. □

Lemma 209 (Class $\mathcal{KL}$ characterization, global case)**.** *The equilibrium x_{eq} is* globally uniformly asymptotically stable *(definition 202) if and only if there is a $\beta : [0,\infty)\times[0,\infty) \to \mathbb{R}$ that is*

1. *zero at the origin: $\beta(0,s) = 0$ for $s \ge 0$;*
2. *jointly continuous on $[0,\infty)\times[0,\infty)$;*
3. *strictly increasing in r for each fixed $s \ge 0$;*

4. *radially unbounded:* $\beta(r,0) \to \infty$ *as* $r \to \infty$*;*

5. *antitone in* s *for each fixed* $r \geq 0$*, with* $\beta(r,s) \to 0$ *as* $s \to \infty$*;*

such that

$$\|\varphi(t) - x_{\mathrm{eq}}\| \leq \beta(\|\varphi(t_0) - x_{\mathrm{eq}}\|,\, t - t_0) \quad \forall\, t \geq t_0 \geq 0,$$

with no restriction on the initial state $\varphi(t_0)$*.*

Condition (4) is what makes the characterization global*: without it* β *is class* $\mathcal{KL}$ *on a bounded radius and the statement collapses to the local one (lemma 208).*

This is the global case of Khalil's Lemma 4.5.

Proof. □

5.7 Lyapunov's theorems for time-varying systems

For a time-varying V the Lie derivative acquires an explicit time term, and positive definiteness of $V(t,\cdot)$ on its own buys nothing uniform in t_0: V could flatten as $t \to \infty$. Sandwiching V between two positive-definite functions of the state alone is what restores uniformity.

Lemma 210 (Chain rule for a time-varying V)**.** *Let* $V : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ *be differentiable and let* φ *be a trajectory of* $\dot{x} = f(t,x)$ *on* $[t_0, \infty)$ *(definition 187). Then for* $t > t_0$*,*

$$\frac{d}{dt}\, V(t, \varphi(t)) = DV(t, \varphi(t))\, [\, (1,\ f(t, \varphi(t)))\,].$$

The first slot of the derivative picks up the explicit time dependence of V*, which is what distinguishes the non-autonomous Lie derivative from the autonomous one.*

Proof. □

Theorem 211 (Lyapunov's uniform stability theorem)**.** *Let* $r > 0$ *and let* $V : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ *be differentiable. Suppose there are* W_1, W_2*, continuous on* $\overline{B}(0,r)$*, vanishing at the origin and strictly positive elsewhere on it, with*

1. $W_1(x) \leq V(t,x) \leq W_2(x)$ *for all* $t \geq 0$ *and* $\|x\| \leq r$*;*

2. $DV(t,x)\, [\, (1, f(t,x))\,] \leq 0$ *for all* $t \geq 0$ *and* $\|x\| \leq r$*.*

Then the origin is uniformly stable (definition 194).

The sandwich is what makes the conclusion uniform in t_0*:* W_1 *and* W_2 *do not depend on* t*, so the* $\delta(\varepsilon)$ *extracted from them does not either.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.8, hypotheses (4.22) and (4.23).*

Proof. Bound W_1 below and W_2 above by class $\mathcal{K}$ functions α_1, α_2 (theorem 186). Fix a level d below both ranges and set $c = \alpha_2^{-1}(d)$, so that starting within c of the origin forces $V(t_0, \varphi(t_0)) < d$. Since V is nonincreasing along the trajectory it stays below d, and $\alpha_1(\|\varphi(t)\|) \leq V \leq d$ confines φ to the ball of radius r — so the estimate never leaves the region where the hypotheses hold. Chaining the sandwich gives $\alpha_1(\|\varphi(t)\|) \leq \alpha_2(\|\varphi(t_0)\|)$, i.e. $\|\varphi(t)\| \leq (\alpha_1^{-1} \circ \alpha_2)(\|\varphi(t_0)\|)$, a class $\mathcal{K}$ bound; conclude by lemma 207. □

Theorem 212 (Lyapunov's uniform asymptotic stability theorem)**.** *In the setting of theorem 211, strengthen the derivative hypothesis to*

$$DV(t,x)\,[\,(1, f(t,x))\,] \le -W_3(x) \qquad \forall\, t \ge 0,\ \|x\| \le r,$$

for a third W_3 continuous on $\overline{B}(0,r)$, vanishing at the origin and strictly positive elsewhere on it. Then the origin is uniformly asymptotically stable (definition 201).

A strictly negative $\dot{V}$ alone would not do: W_3 must be bounded away from zero on each annulus, which is exactly what positive definiteness of a function of x alone buys.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.9, hypothesis (4.24). Khalil's two "moreover" clauses are not part of this conclusion:*

- *the explicit bound $\|x(t)\| \le \beta(\|x(t_0)\|, t - t_0)$, available here via lemma 208;*
- *the global case, when $D = \mathbb{R}^n$ and W_1 is radially unbounded.*

Proof. As in theorem 211, take class $\mathcal{K}$ bounds $\alpha_1, \alpha_2, \alpha_3$ for W_1, W_2, W_3. Then $\dot{V} \le -W_3(x) \le -\alpha_3(\|x\|) \le -\alpha_3(\alpha_2^{-1}(V))$, so $v(t) = V(t, \varphi(t))$ satisfies the scalar differential inequality $D^+v \le -\alpha(v)$ with $\alpha = \alpha_3 \circ \alpha_2^{-1}$ of class $\mathcal{K}$. theorem 239 supplies a class $\mathcal{KL}$ function σ with $v(t) \le \sigma(v(t_0), t - t_0)$. Unwinding the sandwich,

$$\|\varphi(t)\| \le \alpha_1^{-1}(\sigma(\alpha_2(\|\varphi(t_0)\|),\, t - t_0)),$$

and the right-hand side is class $\mathcal{KL}$ in its two arguments; conclude by lemma 208. □

5.8 Linear time-varying systems

Hespanha states Lyapunov stability for the continuous-time linear time-varying system $\dot{x} = A(t)x$ as four conditions on its homogeneous state response $x(t) = \Phi(t, t_0)x_0$. None of them is taken as a definition here: each is proved *equivalent* to the corresponding predicate of the previous sections, so the linear track and the nonlinear track share one notion of stability rather than two.

Two things stand between the textbook conditions and the predicates, and both are settled in this section. The predicates quantify over trajectories where the textbook quantifies over initial conditions, which the trajectory characterization below closes in both directions; and the predicates bound the response from a *ball* where the textbook bounds it from every initial state, which homogeneity closes. The second is why a linear system has no nontrivial region of attraction, recorded as **??**.

5.8.1 The system and its response

5.8.2 Trajectories are state transition responses

5.8.3 Hespanha's Definition 8.1

5.9 Linear time-invariant systems

For constant A the state transition matrix is $e^{A(t-t_0)}$ (theorem 69), and the stability conditions of the previous section collapse to a condition on the eigenvalues of A alone. This is Hespanha's Theorem 8.1.

Clauses (1) and (4) of that theorem — marginal stability and instability — ask that the Jordan blocks at imaginary-axis eigenvalues be 1×1. They are not recorded here: they need a Jordan normal form, which is a project rather than a lemma.

5.9.1 The Hurwitz condition and its rate-indexed refinement

The rate-indexed predicate records a strict spectral margin. It is related to the plain Hurwitz condition by a spectral shift (theorem 29), and the two agree at rate zero.

The Hurwitz condition bounds each eigenvalue away from the imaginary axis one at a time; because the spectrum is compact, that upgrades to a uniform margin (**??**). This is what turns a qualitative hypothesis into a quantitative rate, and it is used twice below.

5.9.2 Equivalent analytic conditions

Nothing in this subsection mentions a system: the Hurwitz condition, decay of e^{At}, and an explicit exponential envelope for $\|e^{At}\|$ are shown to be the same condition on the matrix. Two generic facts about exponential envelopes are recorded first.

5.9.3 Hespanha's Theorem 8.1

5.9.4 The Lyapunov equation

Hespanha's Theorem 8.2 ties the spectral condition to solvability of $PA + A^{\mathsf{T}}P = -Q$. The direction a Lyapunov certificate is *used* in comes first: exhibiting a single positive-definite P proves the spectrum lies in the open left half-plane.

The converse is a construction. One finite block of the integral $\int_0^m e^{tA^{\mathsf{T}}} Q e^{tA}\,\mathrm{d}t$ is summed into a convergent discrete Lyapunov series, with convergence and uniqueness both coming from the contraction $\|e^{Am}\| < 1$. A Hurwitz matrix supplies that contraction, which is theorem 32.

5.9.5 The rate form

The Lyapunov inequality $PA + A^{\mathsf{T}}P \preceq -2cP$ carries an explicit decay rate: along a trajectory it gives $\dot{V} \le -2cV$ for $V = x^{\mathsf{T}}Px$. It is not one of Hespanha's clauses, but it is the shape every contraction and guaranteed-decay-rate condition is written in, so it is recorded as a sixth equivalent.

5.9.6 Theorem 8.2

5.9.7 Quadratic forms along a linear vector field

What makes a solution of the Lyapunov equation a Lyapunov *function*: the forcing matrix is exactly the dissipation rate of the associated quadratic form.

5.9.8 A quadratic certificate for an unstable eigenmode

The instability counterpart (theorem 33): an eigenvalue with positive real part yields a real quadratic Chetaev certificate. The construction solves a shifted Lyapunov–Sylvester equation, whose injectivity rests on a nonresonance condition on the spectrum — which is what the lemmas below supply.

Chapter 6

Class $\mathcal{KL}$ functions

6.1 Inverses of antitone functions

A function W that is continuous and strictly antitone on $(0,\infty)$, blows up at 0^+ and vanishes at $+\infty$ surjects onto $(0,\infty)$, and its right inverse has the same shape. This is the mechanism behind the class $\mathcal{KL}$ construction of the next section: W is a mean-value bound and its inverse supplies the decay rate.

Lemma 213 (Surjectivity onto $(0,\infty)$ from the two boundary limits)**.** *Let $W : \mathbb{R} \to \mathbb{R}$ be continuous on $(0,\infty)$ with $W(\eta) \to +\infty$ as $\eta \to 0^+$ and $W(\eta) \to 0$ as $\eta \to +\infty$. Then for every $s > 0$ there exists $c > 0$ with $W(c) = s$.*

Proof. □

Lemma 214 (The right inverse inverts)**.** *Under the surjectivity conditions of lemma 213, $W(\mathrm{invFunOn}\, W\, (0,\infty)\, s) = s$ for all $s > 0$.*

Proof. □

Lemma 215 (Positivity of the right inverse)**.** *Under the surjectivity conditions of lemma 213,* $\mathrm{invFunOn}\, W\, (0,\infty)\, s > 0$ *for all $s > 0$.*

Proof. □

Lemma 216 (The inverse of a strictly antitone function is strictly antitone)**.** *If W is strictly antitone on $(0,\infty)$, then* $\mathrm{invFunOn}\, W\, (0,\infty)$ *is strictly antitone on $(0,\infty)$.*

Proof. □

Lemma 217 (The inverse vanishes at $+\infty$)**.** *Under the conditions of lemma 213 and with W strictly antitone,* $\mathrm{invFunOn}\, W\, (0,\infty)\, s \to 0$ *as $s \to +\infty$.*

Proof. □

Lemma 218 (Continuity of the inverse)**.** *Under the conditions of lemma 213 and with W strictly antitone,* $\mathrm{invFunOn}\, W\, (0,\infty)$ *is continuous on $(0,\infty)$.*

Proof. □

Lemma 219 (The inverse blows up at 0^+)**.** *Under the conditions of lemma 213 and with W strictly antitone,* invFunOn $W\,(0,\infty)\,s \to +\infty$ *as* $s \to 0^+$.

Proof. □

6.2 Class $\mathcal{KL}$ functions and the Osgood construction

Definition 220 (Class $\mathcal{L}$ function)**.** A *class* $\mathcal{L}$ function on $[0,\infty)$ is continuous, strictly positive, and strictly decreasing, and satisfies $\sigma(s) \to 0$ as $s \to \infty$. It carries the time decay of an asymptotic stability estimate: the product of a class $\mathcal{K}$ function of the initial deviation with a class $\mathcal{L}$ function of elapsed time is a class $\mathcal{KL}$ bound.

Definition 221 (Singular class $\mathcal{L}$ function)**.** A *singular class* $\mathcal{L}$ function on $(0,\infty)$ is continuous, strictly positive and antitone, with $U(s) \to 0$ as $s \to \infty$ and $U(s) \to \infty$ as $s \to 0^+$. It differs from class $\mathcal{L}$ in two ways: the domain excludes 0, where the function is instead required to blow up, and the decrease is only required to be weak. It arises as the inverse of the sliding-window function in the class $\mathcal{KL}$ characterization of asymptotic stability.

Definition 222 (Class $\mathcal{KL}$ function)**.** A *class* $\mathcal{KL}$ function on $[0,a) \times [0,\infty)$ is continuous, class $\mathcal{K}$ in the first argument, and for each fixed $r > 0$ is strictly decreasing and tends to 0 as $s \to \infty$. It arises as the bound $\|x(t)\| \le \beta(\|x_0\|, t)$ in asymptotic stability estimates.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3. Difference: Khalil says $\beta(\cdot, s)$ is class $\mathcal{K}$; here that is spelled out as separate fields, so no class $\mathcal{K}$ instance is required for each s.

Definition 223 (Global class $\mathcal{KL}$ function)**.** A *global class* $\mathcal{KL}$ function is a map $\beta : [0,\infty) \times [0,\infty) \to \mathbb{R}$ that is class $\mathcal{K}_\infty$ in the first argument (continuous, strictly increasing, zero at zero, radially unbounded) and, for each fixed $r \ge 0$, is antitone and tends to 0 as $s \to \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3 with $a = \infty$.

6.2.1 Building class $\mathcal{KL}$ functions

Two constructions and the two composition rules, in the bounded and the global setting. The product $\alpha(r)\gamma(s)$ of a class $\mathcal{K}$ and a class $\mathcal{L}$ function is the easy one; the Sontag-style cap is the one that does real work.

Definition 224 (Class $\mathcal{KL}$ from a class $\mathcal{K}$ times a class $\mathcal{L}$)**.** If α is class $\mathcal{K}$ on $[0,a)$ and γ is class $\mathcal{L}$, then $\beta(r,s) := \alpha(r)\gamma(s)$ is class $\mathcal{KL}$ on $[0,a)$.

The cap is built in the form $\min(\alpha(r), \sqrt{\alpha(r)\,U(s)})$, so the decay of that particular expression is recorded first.

Lemma 225 (Decay of a geometric-mean cap)**.** *Let $c \ge 0$ and let $U : \mathbb{R} \to \mathbb{R}$ satisfy $U(s) \to 0$ as $s \to +\infty$. Then* $\min(c, \sqrt{c\,U(s)}) \to 0$ *as* $s \to +\infty$.

Proof. □

Theorem 226 (Class $\mathcal{KL}$ from a singular class $\mathcal{L}$ cap)**.** *Let α be class $\mathcal{K}$ on $[0,a)$ and let U be a singular class $\mathcal{L}$ function (positive, antitone, $U(s)\to 0$ as $s\to\infty$ and $U(s)\to\infty$ as $s\to 0^+$). Then*

$$\beta(r,s) := \begin{cases} \alpha(r) & s = 0, \\ \min(\alpha(r), \sqrt{\alpha(r)U(s)}) & s > 0, \end{cases}$$

is class $\mathcal{KL}$ on $[0,a)$.

Proof. Every field but continuity is immediate from the corresponding property of α and U, the decay in s being lemma 225. Continuity is checked in three regimes. For $s>0$ the min branch is a composition of continuous maps. At $(0,0)$ the value is squeezed between 0 and $\alpha(r)\to 0$. At $(r,0)$ with $r>0$, pick a buffer radius $r_1\in(r,a)$; since $U(s)\to\infty$ as $s\to 0^+$, on a neighbourhood we have $\alpha(r_1)<U(s)$, hence $\alpha(r')\le U(s)$ and so $\alpha(r')\le\sqrt{\alpha(r')U(s)}$ — the minimum is locally just α, which is continuous. □

Definition 227 (Post-composition with a class $\mathcal{K}$ function)**.** Let β be class $\mathcal{KL}$ on $[0,a)$ and α be class $\mathcal{K}$ on $[0,b)$, and suppose $\beta(r,s)<b$ for every $r\in[0,a)$ and $s\ge 0$. Then $(r,s)\mapsto\alpha(\beta(r,s))$ is class $\mathcal{KL}$ on $[0,a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1). The range hypothesis is additional: α_1 is defined only on $[0,b)$, so β's values must land there, which Khalil leaves implicit.

Definition 228 (Post-composition with a class $\mathcal{K}_\infty$ function)**.** If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}_\infty$, then $(r,s)\mapsto\alpha(\beta(r,s))$ is class $\mathcal{KL}$ on $[0,a)$. This is definition 227 with the range hypothesis discharged by α being defined on all of $[0,\infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet, with α_1 of class $\mathcal{K}_\infty$; no range hypothesis is then needed.

Definition 229 (Pre-composition in the radius argument)**.** If β is class $\mathcal{KL}$ on $[0,b)$ and α is class $\mathcal{K}$ on $[0,a)\to[0,b)$, then $(r,s)\mapsto\beta(\alpha(r),s)$ is class $\mathcal{KL}$ on $[0,a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2).

Definition 230 (Global class $\mathcal{KL}$ from a product)**.** If α is class $\mathcal{K}_\infty$ and γ is class $\mathcal{L}$, then $\beta(r,s):=\alpha(r)\gamma(s)$ is global class $\mathcal{KL}$.

Definition 231 (Post-composition, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s)\mapsto\alpha(\beta(r,s))$ is global class $\mathcal{KL}$. No range hypothesis is needed, since α is defined on all of $[0,\infty)$.

Definition 232 (Pre-composition in the radius argument, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s)\mapsto\beta(\alpha(r),s)$ is global class $\mathcal{KL}$.

Theorem 233 (Capping a class $\mathcal{KL}$ function by a class $\mathcal{K}$ function)**.** *If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}$ on $[0,a)$, then $(r,s)\mapsto\min(\beta(r,s),\alpha(r))$ is class $\mathcal{KL}$ on $[0,a)$.*

Capping a class $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class. Its use is to recover a range condition: a bound obtained only as a majorant — from a smoothing construction, say — carries no upper control, and capping restores it without disturbing the bound, since the minimum still dominates whatever β dominated.

Proof. Each field is inherited. The minimum of two strictly increasing functions of r is strictly increasing; α does not depend on s, so antitonicity in s survives the minimum; and $\min(\beta(r,s),\alpha(r)) \to \min(0,\alpha(r)) = 0$ as $s \to \infty$ because $\alpha \geq 0$. □

Definition 234 (Capping a global class $\mathcal{KL}$ function)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \min(\beta(r,s),\alpha(r))$ is global class $\mathcal{KL}$ — capping a $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class.

6.2.2 Smoothing a bound into a class $\mathcal{KL}$ function

A candidate bound that has the right qualitative behaviour — monotone in the initial deviation, decaying in time — need not itself be class $\mathcal{KL}$: it may fail continuity or strict monotonicity. These two results replace such a candidate by a genuine class $\mathcal{KL}$ function dominating it, which is how the stability characterizations produce their bounds. The first lemma below is the continuity criterion the construction rests on.

Lemma 235 (Class $\mathcal{KL}$ majorant)**.** *Let $\psi : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ satisfy, for all $r, s \geq 0$:*

1. *$\psi(r,s) \geq 0$;*
2. *$\psi(0,s) = 0$;*
3. *$r \mapsto \psi(r,s)$ is monotone on $[0,\infty)$;*
4. *$s \mapsto \psi(r,s)$ is antitone on $[0,\infty)$;*
5. *$\psi(r,s) \to 0$ as $s \to \infty$;*
6. *$r \mapsto \psi(r,0)$ is continuous at 0 within $[0,\infty)$.*

Then there is a global class $\mathcal{KL}$ function β with $\psi(r,s) \leq \beta(r,s)$ for all $r, s \geq 0$.

Proof. The two hypotheses of **??** are supplied in turn. For uniform convergence, $\psi(r,\cdot) \to 0$ gives a time T beyond which $\psi(r,t) < \varepsilon$, and monotonicity in the radius extends that bound to every $s \leq r$. For uniform stability, continuity of $\psi(\cdot,0)$ at 0 gives a δ with $\psi(s,0) \leq \varepsilon$ for $s \leq \delta/2$, and antitonicity in time propagates it to all $t \geq 0$. □

Lemma 236 (Global class $\mathcal{KL}$ from a decay family)**.** *Let α be class $\mathcal{K}_\infty$ and let U be a family of functions such that*

1. *$U(r,s) > 0$ for all $r > 0$ and $s > 0$;*
2. *$U(r,\cdot)$ is antitone on $(0,\infty)$ for each $r > 0$;*
3. *$U(r,s) \to 0$ as $s \to \infty$, for each $r > 0$;*
4. *$r \mapsto U(r+1,s)$ is monotone on $[0,\infty)$ for each $s > 0$.*

Then there is a global class $\mathcal{KL}$ function β such that

1. *$\beta(r,0) \to \infty$ as $r \to \infty$;*
2. *$\alpha(r) \leq \beta(r,0)$ for all $r \geq 0$;*

3. $\min(\alpha(r), \sqrt{\alpha(r)U(r+1,s)}) \le \beta(r,s)$ *for all* $r \ge 0$ *and* $s > 0$.

Proof. Take $\psi(r,s) := \alpha(r)$ for $s = 0$ and $\min(\alpha(r), \sqrt{\alpha(r)U(r+1,s)})$ for $s > 0$, the radius being shifted by 1 so that U is only ever evaluated at a strictly positive radius. Each hypothesis of lemma 235 holds for ψ: monotonicity in r from that of α and of $r \mapsto U(r+1,s)$, antitonicity in s from that of $U(r+1,\cdot)$, decay from lemma 225, and continuity at $r = 0$ because $\psi(\cdot,0) = \alpha$. The resulting β dominates ψ, which gives the three conclusions; $\beta(r,0) \ge \alpha(r) \to \infty$ gives the first. No property of ψ beyond these bounds appears in the statement. □

6.2.3 The decay ODE and the comparison bound

A class $\mathcal{K}$ decay rate produces a class $\mathcal{KL}$ bound. This is the step that turns a Lyapunov estimate — which controls only the *rate* of decrease — into an asymptotic stability statement, and it is where the scalar ODE $\dot{y} = -\alpha(y)$ enters.

Theorem 237 (The decay ODE has a class $\mathcal{KL}$ solution operator)**.** *Let* α *be class* $\mathcal{K}$ *on* $[0,a)$ *and at most linear near the origin,* $\alpha(x) \le Lx$ *on* $(0, \text{base}]$. *Then there is a class* $\mathcal{KL}$ *function* σ *(definition 222) with* $\sigma(0,s) = 0$, $\sigma(r,0) = r$, *and such that for each* r *the map* $s \mapsto \sigma(r,s)$ *solves*

$$\dot{y} = -\alpha(y), \qquad y(0) = r.$$

That is: the decay ODE's solution operator is itself a class $\mathcal{KL}$ *function.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.4. Differences:*

- *Khalil assumes* α *locally Lipschitz; here the hypothesis is the linear bound* $\alpha(x) \le Lx$ *near the origin, which is what the construction uses.*
- *Uniqueness of the solution is not asserted, only existence of the class* $\mathcal{KL}$ *operator* σ.

Proof. **Osgood's construction.** Let $\eta(y) = -\int_{\text{base}}^{y} \mathrm{d}x/\alpha(x)$ be the time for the solution started at base to reach y. Since $\alpha > 0$ on $(0,a)$, η is continuous and strictly decreasing there; and since $\alpha(x) \le Lx$ near 0,

$$\eta(y) \;\ge\; \int_{y}^{\text{base}} \frac{\mathrm{d}x}{Lx} \;=\; \frac{\log(\text{base}) - \log y}{L} \;\longrightarrow\; +\infty \qquad (y \to 0^{+}),$$

so the origin is not reached in finite time and every value above $\eta(r)$ is attained. Setting $\sigma(r,s) := \eta^{-1}(\eta(r) + s)$, extended by $\sigma(0,s) = 0$, advancing time by s moves η up by s and hence moves the state down; the class $\mathcal{KL}$ conditions follow from monotonicity and continuity of η^{-1}, and differentiating $\eta(\sigma) = \eta(r) + s$ gives $\dot{\sigma} = -\alpha(\sigma)$.

The construction is local to this proof: η and η^{-1} are steps in building a solution, not results about class $\mathcal{K}$ functions. □

Lemma 238 (Lipschitz class $\mathcal{K}$ minorant)**.** ***Assumed without proof.*** *Let* α *be a class* $\mathcal{K}$ *function on* $[0,a)$ *and let* base $\in (0,a)$. *Then there are a class* $\mathcal{K}$ *function* β *on* $[0,a)$, *a function* $\beta_{\text{ext}} : \mathbb{R} \to \mathbb{R}$ *and a constant* $L > 0$ *such that* $\beta(x) \le \alpha(x)$ *and* $\beta_{\text{ext}}(x) = \beta(x)$ *for* $x \in [0,a)$, $\beta(x) \le Lx$ *for* $x \in (0, \text{base}]$, *and* β_{ext} *is continuous and* L*-Lipschitz on all of* $\mathbb{R}$.

Theorem 239 (Comparison bound from a Dini decay condition)**.** *Let α be class $\mathcal{K}$ on $[0, a)$. Then there is a class $\mathcal{KL}$ function σ with $\sigma(r, 0) \leq r$ such that: whenever v is continuous on $[t_0, t]$, takes values in $[0, a)$ there, has bounded forward difference quotients, and satisfies the Dini inequality*

$$D^+v(s) \;\leq\; -\alpha(v(s)),$$

it obeys $v(t) \leq \sigma(v(t_0),\, t - t_0)$.

A differential inequality *with a class $\mathcal{K}$ decay rate therefore yields a class $\mathcal{KL}$ bound — which is what converts a Lyapunov decay estimate into an asymptotic stability statement. No hypothesis is placed on α beyond being class $\mathcal{K}$: the linear-growth condition the construction needs is obtained internally by passing to a Lipschitz minorant.*

Proof. Replace α by a class $\mathcal{K}$ minorant $\beta \leq \alpha$ that is Lipschitz near the origin; this only weakens the Dini hypothesis, and supplies the linear bound $\beta(x) \leq Lx$ that theorem 237 requires. That theorem gives a class $\mathcal{KL}$ σ whose sections solve $\dot{y} = -\beta(y)$ with $\sigma(r, 0) = r$. Since v is a Dini subsolution of the same equation and starts at $\sigma(v(t_0), 0)$, theorem 122 bounds v by it — the perturbed solutions that lemma requires coming from theorem 118, whose hypotheses hold because β's Lipschitz extension is globally Lipschitz. □